Chapter 1: Introduction

The project Web-Based Ticket System for ChainTicket will be briefly described in this chapter, along with the background study, problem statement, project objective, scope, and thesis organization.

1.1 Background Research

The advent of the digital world led to substantial growth in the live entertainment and event industries. Because of this, selling event tickets online has grown to be one of the most significant aspects of our social and economic structure. In the meantime, the current ticketing system is dealing with a number of systemic issues brought on by unequal access to technology, or the "digital divide," and a lack of access to ticket sales for all customers. These issues are related to the difficulties in achieving the Sustainable Development Goal of the United Nations, which is to reduce inequality.

Because scalpers are technologically savvy and take advantage of multiple account, automated ticket purchasing bots, they routinely deny regular customers the opportunity to purchase tickets to shows and other events (Queue-it., 2025). They do this by taking over 40 percent of every primary ticket sale through automated bots that give them power that is not available to most of the public (Imperva., n.d.). Customers who wish to go to public events have two options: either they pay exorbitant prices for tickets in the secondary market, or they completely avoid going to cultural events.

The continued availability of these ticket hoarders using automated systems has led to an increase in the number of unregulated counterfeit tickets in the marketplace that jeopardize public safety, in addition to the individual abuse resulting from the financial exploitation of consumers. When innocent customers buy used tickets, they frequently become victims of different kinds of ticket fraud, such as when dishonest people make duplicate tickets. The victims of ticket fraud won't be aware that they have been duped until they get to the event location because traditional ticketing systems lack a decentralized validation mechanism. Customers will lose faith in the entire digital ticketing sector because of this lack of trust, which will also lead to a great deal of crowd entry friction and numerous operational problems for the promoter.

The core concept of architecture within a blockchain is represented as a Distributed Ledger Technology (DLT) that can be thought of as a decentralized, immutable, and peer-to-peer distributed ledger that stores transaction data chronologically across a distributed group of independent computers (S.A., 2024). Public blockchains (an example being Gnosis Chain which is compatible with Ethereum) do not require validation of a single central authoritative entity for state transitions but rather utilize a decentralized consensus to determine their operational state. When a transaction occurs on the network, it is grouped with other concurrent transactions by a cryptographically secured structure called a block. This block is then broadcast across thousands of validating nodes situated globally where

algorithmically determined consensus protocols will verify the structural validity of the transactions in the block prior to appending it to the historical ledger as a new state. By superseding traditional systems of centralized trust with the introduction of dynamic mathematically determinable consensus systems the public ledger framework provides a means for utilizing trust as the foundation for executing data.

The fundamental security framework of blockchain is based on mixing cryptographic hashing with asymmetric crypto and decentralized architectural resilience which creates a mathematically unalterable source of truth for the data stored within the blockchain ledger. Every independent block has its own unique cryptographic identifier or also call as "hash value" which is computed through secure hashing algorithms and serves as a digital fingerprint for the dataset contained in that block. Each new block will also contain the actual cryptographic hash of the previous block in the header of the block, thus establishing a continuous and interconnected cryptographic link between each block within the blockchain.

From a software engineering viewpoint, this design helps to ensure the integrity of the data being stored because any malicious actor trying to change or falsify the historical ticket transaction data stored in an older block would immediately cause the cryptographic hash of that older block to become invalid (S.A., 2024). All dependent subsequent blocks would then fail validation as a result and be automatically rejected by the validator nodes of the network. In addition, blockchain technology provides multiple identical copies of the blockchain ledger that are synchronized in real-time and distributed across an open, global network of validators, creating an environment where there is absolutely no single point of failure within the network, which protects the digital assets of the network from malicious database injection attacks, unauthorized administrative overrides, and large-scale distributed denial-of-service (DDoS) attacks.

Besides, there is an urgent need for technological innovation in the scope of national digital infrastructures due to the underlying structural crisis as part of Sustainable Development Goal (SDG) 9, which is focused on Industry, Innovation and Infrastructure. The traditional centralized ticketing database is fundamentally opaque, leaving the entire lifecycle of a ticket beyond the purview of the public view once it has been transferred from the original primary distributor. Therefore, there is no way for consumers to verify the fairness of the allocation of tickets and lots of potential for insider manipulation of allocations and for institutional oversells that violate basic purchaser rights. Current structures of monopolistic imbalance in ticketing marketplace do not create a balanced and self-regulating ecosystem that safeguards against violations of intellectual property rights, violations to public order, or economic injustices to the other members of society.

In order to overcome traditional barriers, societal and infrastructural, this project suggest an entirely new way to transact based on a paradigm shift from vulnerable centralised ticketing database to a decentralised ledger architecture using blockchain technology. By moving the governance of ticket

transaction and validation to a blockchain framework, we can create a transparent and immutable record of the entire lifecycle of an event ticket, from minting and primary sale through to secondary transfer. This unprecedented level of transparency allows everyone involved in the purchase and transfer of tickets to have independent access to audit the original source of tickets, track the movement of tickets throughout their lifecycle, and verify their ownership history. This eliminates the opaque nature of traditional ticketing monopolies and protects against systemic distribution of tickets.

The technical intervention of this project is use of non-fungible tokens (NFTs) and automated smart contracts that convert digital tickets from easily replicated text files into non-replicable cryptographic assets created on the blockchain to become unique and irrefutable proof of ownership, each with an ownership history that cannot be altered or duplicated by any unauthorized actor (Aswal P., 2026). Smart contracts function as decentralized execution platforms to enforce the rules of a marketplace mathematically, without the oversight of a human intermediary. Although transacting on a public blockchain incurs a "gas fee" for all transactions, this fee is well worth the cost given that users get significant protection against fraud and price markdowns on black markets due to the degree of security inherent in using a blockchain to create tickets. Therefore, the gas fees incurred for every transaction on a blockchain are justified as an appropriate warning system designed to protect users from lost funds and protect the overall integrity of the operating system which is the blockchain.

As such, this decentralised model creates an environment which allows for the win-win situation of the event organiser, the initial buyer and the secondary trader as opposed to an enforced ban on peer-to-peer trading which drives consumers away from safe and legal markets, to dangerous and illegal black markets by way of the event organisers' restrictive and ineffective policies. This blockchain network will provide a highly secure, native, peer to peer environment through which participants can engage in voluntary economic transactions. All activity on this network will fall under the automated-state transitional model of architectural design whereby all participants are guaranteed to operate according to pre-existing consensus-based rules; thus, ensuring that secondary market activities will be conducted in a safe, transparent manner and in compliance with ethical commerce.

ChainTicket is a decentralized, web-based ticketing system which was created to address the many systemic flaws that exist in the current ticketing industry, including those that have been identified as the primary vectors driving up the cost of tickets in an open market such like against the will of both artists and fans. A significant portion of those systemic flaws are likely related to the rapid growth of the internet. ChainTicket's focus will be addressing the issues of scalping and the digital divide by implementing an off-the-shelf, smart contract-based secondary market that legally enforces a strict price limit for resale tickets to maintain an economically safe resale value for all parties involved in a given transaction. ChainTicket will enforce those values through a smart contract mechanism that limits resale ticketing to a percentage of that ticket's original face value, thereby negating the speculative profit

margin of both bots buyers and scalpers. In addition, an embedded royalty fee distribution mechanism will ensure that a percentage of the value of each transaction that occurs on the secondary market is distributed to and benefits to the event organizer. This dual-purpose mechanism will provide an economically sound compromise between artists and fans while also deterring speculative purchases and controlling the resale distribution of an event.

ChainTicket intends to provide a concrete solution for tackling the extensive problem of ticket fraud and duplication through the implementation of an end-to-end secure verification framework that consists of a consumer facing web portal and an operable verification terminal that will verify the authenticity of a ticket prior to entry into the event. Instead of utilizing static assets, which are inherently insecure and can be duplicated, the verification system will utilize a dynamically generated QR code that is connected in real time via blockchain state data. The consumer facing web portal will calculate and refresh a time-sensitive cryptographically secure token on the user's mobile device, which can only be decrypted by the verification terminal at the event venue. Because the dynamic QR code will update frequently, and the previous static version will become invalid once either the QR code has been duplicated or a screenshot has been taken, there are inherent protections put in place to prevent a bad actor from repurchasing the same ticket multiple times and giving a second hand purchaser assurance that the ticket is 100% legitimate.

In conclusion, ChainTicket provides transparent systems and flexible circulation capabilities through optimising their operational environment to only permit the sale of tickets for events to ensure maximum throughput of transactions while incurring minimum gas fees incurred for these transactions which is cost of using the public ledger. Additionally, ChainTicket enables individuals to have full rights to directly monitor their tickets via the state variables of the public ledger, which provides complete assurance of creating complete trust and eliminating the possibility that a vendor has oversold inventory. Finally, by replacing arbitrary and centralised vendor's policies with immutable policies on the blockchain, which are enforced by the blockchain, ChainTicket will change the hostile and high-risk environment of purchasing tickets into a fair, inclusive and extremely resilient method of digital infrastructure for modern society.

1.2 Problem Statements

I. Scalper problem which about algorithmic asymmetry and centralized account abuse

The mass-scale hoarding of event tickets is not mitigated on centralized ticketing sites, such as AXS, Ticketmaster, and Live Nation. Automated bots and dishonest brokers are able to successfully bypass structural limitations placed on ticket purchases using high-powered computing systems, false IP addresses, and fictitious identities. Additionally, many of the centralized ticketing sites often ignore these practices because the secondary flips of those tickets create a source of revenue through multi-tiered transactional fees. This leaves the legitimate consumer out of pocket and often purchasing tickets for resale at markups of 1,000% or more (Abramova A., 2025), thus disrupting the equality of the marketplace and infringing upon the consumer's rights as a purchaser of a product.

ChainTicket uses a multi-level identity & access mitigation protocol to break the automation loop at the entry point. The added verification of identity through standard reCAPTCHA and a strong compliance concept based on KYC, attempts to disallow anyone using disposable virtual credit cards from purchase event ticket. This creates additional expenses for Sybil attack bots to operate. The smart contract behind the ticket NFT will not allow the gift of the ticket NFT non-custodially so the ticket NFT can only be transferred via the platform's official escrow ecosystem, thereby preventing automated external scalping scripts from operating.

II. Fraud Issue about duplication, speculative prelisting, and gate verification failure

Digital ticket representations such as static barcodes and unsecured PDF-files are susceptible to cloning and fraudulent duplication, primarily because they contain no public, real-time cryptographic validation ledger to verify their authenticity. In addition, a lack of an immediate, public verification system for ticket authenticity allows bad actors to consistently commit double-sell fraud by posting the same ticket barcode on multiple secondary market-selling platforms and social networks at the exact same time. As reported by CCID director Comm Datuk Rusdi Mohd Isa, there has been a surge in reported concert ticket frauds over the past few months resulting from widespread use of digital ticketing and unauthorized duplication, hundreds of local fans were reportedly defrauded of around RM96,000 (Tee K., 2026), upon discovering at the entrance gates of the venue that they possessed invalid or duplicated tickets. The sale of prelisted (ticket brokers' guarantees that they will deliver the tickets to fans) tickets creates an artificially volatile and risk-prone environment for fan behavior.

The ChainTicket technology solves the replication problem at a mathematical level by relocating the ticketing infrastructure to an immutable blockchain ledger. Each ticket becomes a unique non-fungible token (NFT) with a tamper-proof ownership history. The web application on the client side hides the ticket data completely until two hours before an event; at that point, it activates a Dynamic QR Code Engine. Every 30 seconds, the Dynamic QR Code Engine computes a new digital token based

on live blockchain state and Unix time stamp. The event organizer's verification terminal scans the code presented at the venue; the verification terminal checks to confirm the ticket has been freshened up locally and then signs the state change using cryptography to allow access, making it impossible to duplicate static copies of tickets or sell speculative copies prior to their availability for resale.

III. Transparency Issue such like monopolistic intermediaries and invisible secondary arbitrage

Legacy ticketing systems are primarily opaque, using centralised databases that have concealed all internal ticket allocation, distribution and transaction lifecycles from the public. Ticket brokers and other intermediaries have taken advantage of the architectural opaqueness of these systems, using them to execute unregulated arbitrage pricing; allowing unchecked speculative trading without a standardised and regulated audit trail. For example, many extremely sought after concert tickets are flipped multiple times through privileged insider networks before the start of the event; the intermediary platforms collect trifecta commissions during this entire process (Seglins D., 2025), while many original artists and actual ticket buyers suffer the financial consequences. The consumer does not possess the right to audit the equity of the ticket distributions or verify that the intermediary's operating practices qualify as illegal institutional overselling.

ChainTicket re-establishes public transparency by recording every occurrence throughout all stages of the ticket life cycle from creation through trade on the open-access and distributed public ledger, thus satisfying SDG 9 and 10 infrastructure vision. In place of centralized databases used in current systems, ChainTicket replaces them with immutable, self-executing smart contracts. This open data continues to allow all users to track exactly where tickets have gone, creating complete structural equality, while also preventing any ticket issuer or secondary market from hiding any hidden costs, price discrimination, or deceptive pricing models.

IV. Circulation issue because absence of regulated marketplaces and extreme speculative exploitation

Automated regulation does not exist in the secondary marketplace for tickets. Therefore, many tickets that are sold on the secondary market related to high-profile tours are sold at prices that are extremely excessive in relation to their face values, such as in some instances 70 times greater than their face value (Abramova A., 2025). When primary-market ticket sellers attempt blanket prohibitions against the transferability of tickets to combat scalping, such strict enforcement actions result in fans that find themselves in desperate situations turning to completely unregulated black market sellers or social media platforms (King A., 2025), both of which present vastly higher levels of risk for consumers to become victims of fraud. The underlying problem is that there is no reasonable, rule-based, framework in the peer-to-peer marketplace for legitimate buyers who cannot attend an event to locate

and purchase tickets, while also providing adequate protections for the marketplace as a whole from becoming an investment vehicle for speculating and profiting.

ChainTicket creates a safe, programmatic peer-to-peer environment for ticketing, through its smart contracts, as an alternative to ticket reselling. It has mathematically locked ticket reselling to a price ceiling of 10% more or less than the original ticket's value, therefore leaving no room for speculative profit. The ticketing application uses an aggregated blind public pool method for ticket listing, preventing the public from being able to identify the price of an individual ticket and only letting them know the maximum and minimum statistical prices at any given time. In addition, it has built an automated royalty fee system that charges a royalty fee to an event organizer whenever a ticket is resold. The sale of a ticket on the secondary market now gets rid of the dependent on black-market profitability and creates a completely compliant and sustainable secondary ticket use case.

1.3 Objectives

The objective are:

I. To analyse the current issue faced by ticket buyer in ticket scalping and fraud issue that will impact buyer's user experience in term of ticket rush and second-hand market for tickets.

II. To design the ticket system integrate with block chain element and ticket verification terminal feature for ChainTicket.

III. To develop a Web-Based for concert ticketing system integrate with the block chain ticket create, trade and verify features of the ticket platform.

IV. To evaluate the reliability and security of concert ticketing system integrate with the block chain ticket create, trade and verify features of the ticket platform.

1.4 Scope

I. User Scope

- The IT literacy will be the main focus in user scope because ensure the designed UI are usability-friendly for user at all level and meet the information security level

II. System Scope

- The system will develop integrate with block chain to improve the transparency and fraud prevention of ticket system by storing and processing ticketing information on the blockchain.
- Service will only focus on event ticket due to block chain need pay gas fee for process the transaction.

1.5 Thesis Organization

Figure 1 will show about the Thesis Organization of whole report and below will detail description of the thesis structure in whole report.

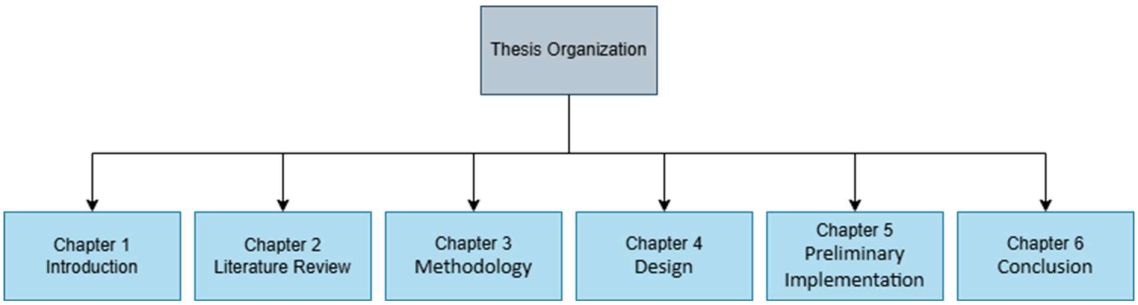


Figure 1 Thesis Organization

In Chapter One, the issue is briefly discussed, along with the system's overview, background information, problem statement, functionality, and possible user. The chapter outlines and emphasizes the issue that people are currently facing. The problem should be resolved, and the user should gain from the present development. Additionally, the chapter outlines the user scope, system scope, and system objective. This chapter aids in defining the boundaries of the system to be constructed and analyzing the project's objectives.

The project's basic knowledge and understanding of the current system are provided in Chapter Two. This chapter is crucial for the project to comprehend the necessary skills and functionality as well as the advantages and disadvantages of the current system. The analysis will aid in the creation of a new system that is improved and has features not found in the current system. The results of the literature review also include information on the databases and frameworks used to create Web-based applications.

Chapter Three describes the project's methodology, including how the software process model was chosen, how it was introduced, and how each stage of the model was explained. The timeline for FYP1 will be presented, along with the project framework, software, and hardware requirements.

The system diagram, which serves as a development guide, is presented in Chapter Four. Before the development stage, the various design diagrams and system architecture will give an understanding of the behavior and structure of the system.

The system prototype, which demonstrates how the system functions and how users interact with it, is presented in Chapter Five. Every interface will be shown, along with an explanation of each function.

The project concludes in Chapter Six, which documents all data and findings to identify system limitations and carry out system quality improvement and enhancement.

Chapter 2: Literature Review

2.1 Introduction

Chapter 2 will describe concepts and technologies used in the project. This will include an overview of blockchain technology, traditional event ticketing systems, and fraud. Section 2.2 will descript about e-ticketing and its challenge in now. Section 2.3 and Section 2.4 discuss about the blockchain technology and its characteristics that will be use integrate in ticketing system. Section 2.5 and Section 2.6 explain about the QR code authentication and the technology that related with this project. Section 2.7 and 2.8 descript about existing system and compare between each, Section 2.9 and 2.10 discuss about the framework and database that will use in this project.

2.2 Electronic Ticketing

2.2.1 What is e-ticketing

An e-ticket is basically a digital version of a paper ticket. Instead of printing and delivering the ticket, the digital ticket is automatically created and sent to the person using their email or account. The switch from physical tickets to e-tickets was caused by the practical issues associated with using real paper. Physical tickets can be lost, ripped or ruined, and they can be faked very easily. In addition, paper tickets create problems for the persons organizing the event, example like ticket printing costs, slow check-in process and production of waste from the tickets (MeTicket, 2026).

The movement towards e-ticketing has been greatly facilitated by the improvement of mobile computing and scanning systems. The platforms that deal with e-ticketing issue unique QR codes with each ticket, which link to the account of the buyer. The QR code use implies that the event staff can validate the tickets immediately by means of their mobile devices.

Because of the flexibility and security of e-ticket system, it has been adapted across in that industry to streamline operations:

I. Academic institutions utilize e-tickets for university workshops, student orientations, and graduation ceremonies to efficiently manage student attendance and venue capacity.
II. Large-scale international conferences, such as the Singapore FinTech Festival, rely on e-tickets to quickly process thousands of industry professionals, often integrating the ticket with smart lanyards or networking apps.
III. The commercial sector, particularly transportation, heavily utilizes e-ticketing for bus, train, and airline boarding passes to provide passengers with seamless, self-service travel experiences.

2.2.2 Conventional e-ticket process

The process of ticketing for events has dramatically changed. Previously, the workflow for ticketing including physical logistics as tickets were either collected in person or printed from home

creating the risk of loss and damage of paper tickets for buyers. For event organizers the process bought printing costs, shipping costs, and box office issues. When entering ticket holders manually checked against the written lists causing delays and long queues.

The conventional process from MeTicket (2026) about physical paper to e-ticket process by shifting the whole lifecycle of the ticket into digital platform and mobile device:

Step 1: When a user purchases a ticket online, the platform's backend immediately generates an electronic ticket. Instead of requiring physical delivery or pickup, this digital pass is delivered instantly to the buyer via email or directly into their personal user account.

Step 2: The e-ticket is stored securely on the user's smartphone. Each ticket is embedded with a unique QR code tied specifically to the buyer's account, alongside essential event information such as the venue, date, and ticket type. This completely mitigates the risk of the ticket being physically lost or damaged.

Step 3: Upon arriving at the event, the attendee bypasses manual name checks. They simply open the e-ticket on their mobile device and present the unique QR code to the event staff.

Step 4: Event personnel utilize dedicated mobile scanner applications to scan the presented QR codes. The digital system instantly authenticates the code against the central database to confirm whether the ticket is valid, if it is a duplicate, or if it has already been scanned. This automated validation replaces slow manual checks with fast scanning, significantly accelerating the entry process while tracking attendance in real time.

2.2.3 Challenge of existing e-ticketing

Although standard e-ticketing has been successful in addressing several limitations related to physical tickets, the use of centralized Web2.0 architecture and static digital formats has created numerous problems in terms of security and economy for the industry. In particular, there are the following difficulties related to this practice:

I. Standard e-tickets are often presented in the form of a static image such as a QR code or PDF. Static digital information makes it easy to screenshot and distribute tickets illegally.

II. The possibility of duplication means that fraudsters can sell one ticket to various buyers. The centralized systems that are in use today operate on a “first to scan” principle which means that only the first buyer will be able to enter the venue while other legitimate customers will not access the venue leading to numerous disputes.

III. The existing e-ticketing systems are deficient in authority and control in the secondary market. Scammers use automated software, known as Bots, to artificially buy tickets in bulk when they first become available. After that, the tickets are held for resale through unauthorized third-party platforms for price markups. Since traditional Web2.0 databases do not have

programmable rules to regulate post-purchase transactions, ticket sellers cannot impose limitations on maximum prices or stop ticket scalpers from cheating.

2.3 Blockchain Technology

2.3.1 Definition of blockchain

Essentially speaking, a blockchain is a digital ledger or decentralized database that is shared across the nodes in a network. It keeps data in a secure way, while being transparent and durable to manipulation (Hayes, 2026). The technology is based on a number of structural elements:

I. Based on the blockchain systems, it is possible to have the distributed system instead of the traditional centralized database that forces the user to depend only on the one center where information can be maintained. All information consists of data that is placed into blocks and should be kept by every network's user. In order to use the information, the copies of the data should be identical.

II. The most important feature of public blockchain systems is that it possesses no owner of information; all people that are using this system bear responsibility for it. Thanks to the distribution of the data between numerous nodes of the network, there can be no central point of error.

III. Blockchain systems ensure that there can be no change in the data after it enters the system. The safety is achieved with the help of cryptography. The original hash is generated by every block while keeping the hash of the block before, which allows linking the blocks in the chronological line. That is why any bad action will lead to the alteration of the initial hash, which will not be similar to any other in the chain.

2.3.1.1 Type of blockchain

Depending on network access rules and governance structures from Cyfrin Team (2024), blockchains can be categorized into four primary types:

Type of blockchain	Description	Example chain
Public Blockchains	• These are entirely decentralized and permissionless networks accessible to anyone with an internet connection • The proposed system selects a Public Blockchain for its core transactional layer to ensure transparent, trustless, and publicly verifiable NFT ticket ownership, effectively preventing duplication.	Bitcoin Chain

Private Blockchain	• These are permissioned networks designed specifically for enterprise needs and are only accessible to authorized individuals.	Corda Chain
Hybrid Blockchains	• These architectures are designed to combine the most advantageous features of both public and private networks. • To overcome the scalability limits of a purely public network, this project implements a hybrid Web2.5 approach, efficiently managing heavy media data off-chain while securing core ticketing logic on-chain.	Dragonchain
Consortium Blockchains	• These are permissioned networks governed and managed collaboratively by a consortium of multiple organizations rather than a single entity.	R3 Corda, Hyperledger Fabric

Table 2.1 Type of Blockchain

2.3.2 Blockchain architecture

A blockchain is a decentralized way for storing transactional records on a public ledger. The way that each transaction is verified is through a consensus process where all participants vote on whether a particular transaction should be added to the ledger. This is done without dependence on any central authority. The architecture of a blockchain is structured in several layers: The Data Layer, Network Layer, Consensus Layer, Smart Contracts Layer, Application Layer, and Incentive Layer (Wang X.Y., 2024). The "immortal" nature of the blockchain is created using Mathematical Hash Functions specifically SHA-256 is hash algorithm that using by Bitcoin chain (Razi, et al., 2023), which take an arbitrary piece of information call as “input” and then create a unique cryptographic code call as "output" representing that input. Hash Functions are one-way only so there is no way to figure out the original piece of information from the unique hash code. Therefore, once a transaction has been written to the blockchain, it cannot be deleted, modified, or reverted back to its previous state.

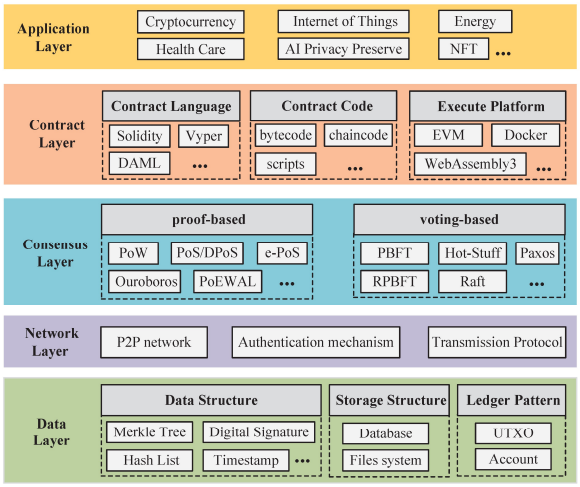


Figure 2.1 Layer of Blockchain

These sequentially cryptographic links create a system that makes it nearly impossible to tamper with records or systems. Blocks are linked to one another using sequential hash pointers means the hashes associated with the contents of their predecessor. If any change is made to any historical block's contents or data, then the associated hash pointer becomes invalid whenever they point at defective hashes, which causes all subsequent blocks that reference and chained to this block will become invalid, enabling immediate detection of problems (Razi, et al., 2023). Additional confidence in the traceability of transactions is assured by using a distribution model that creates a permanent record of every transaction that has occurred. These design elements create an immutable chain of historic execution paths, therefore eliminating dependence on a central authority for control and decreasing the magnitude of risk associated with a single point of failure.

Elliptic Curve Cryptography (ECC) represents the surety behind user identities and user assets with an asymmetric encryption algorithm for the cryptographic framework being used. ECC uses a secure key pair (private key, k; a public key, Q): k is an integer; Q is determined by the equation Q = kP (Razi, et al., 2023), where P is a designated base point on an elliptic curve. The elliptic curve discrete logarithm problem is extremely difficult; therefore, it is computationally infeasible for a malicious user to derive the private key from the other two coordinates. This design enables a user to transact without revealing his or her identity in that a user will digitally sign a transact using a new virtual or email-like address to conduct the transaction and therefore he or she will not be able to identify by an actual located identifier or some other means via actual identifier.

2.4 Characteristic of blockchain

The structural integrity and operational reliability of blockchain technology are derived from three fundamental characteristics from Hayes A. (2026):

I. Blockchain networks are characterized as distributed ledgers in which transactions can be observed by users in the network. In public networks, it is possible for each node in the network to see what has happened on the blockchain using blockchain explorers, but since the transaction data on the blockchain is encrypted, users are generally assured of their anonymity. Nevertheless, even though the data in the ledger is encrypted, it still retains its transparency in terms of the wallet addresses used by the parties involved in the transaction.
II. Data inputted and verified gets entered in the blockchain, becoming irreversible and non-editable. The reason behind such immutability is that all blocks in the chain are linked to one another through the use of a unique, encrypted hash code. Thus, if one attempts to change a block from the past, it will not work as it will not hold a compatible hash code with its parent block.
III. Blockchain technology secures information without using intermediaries throughout vast networks and offers opportunities to control and safeguard information on this network. Since the information belongs to many users, all the changes must be confirmed by all participants of the networks at the

same time and in the same manner, and no single user can change the information independently. The need for the cryptographic proof makes it hard for criminals who may want to break the chain and falsify information, as they would need to control a significant part of the network.

2.5 QR code authentication

QR code verification can be regarded as an approach to security that employs a readable code along with codes for accessing a person's identity or enabling access into a place, system or account. The conventional sequence of events in QR code authentication is to redirect the identity verification procedure to a reliable smartphone or alternative secure application (Baugham, 2026). The authentication process entails a four-pronged system of implementation of QR codes:

I. The system generates a temporary, unique QR Code specifically for a single session or login attempt.
II. The user scans this code using a trusted mobile device or a dedicated authenticator app.
III. The system evaluates the request by checking expiration limits, session matching, and potentially requiring additional proof such as biometrics, device trust, or one-time codes.
IV. Based on the verification results, the system either grants access, blocks the request, or prompts the user with the next required security step.

Importantly, QR Code does not hold any permanent reusable credentials. It only holds a session reference or secure token request or short-term user authentication information allowing entry to an authorization process. In order to incorporate strong security, good implementations use dynamic QR Codes with short expiration periods plus trusted device checks and constant session monitoring. Below is the compare between Static and Dynamic QR code:

Feature	Static QR Code	Dynamic QR Code
Data Encoding	Destination data is directly and permanently encoded into the pattern.	Encodes a short redirect URL, which directs the scanner to an intermediary server.
Editability	Fixed; the destination or content cannot be altered once generated without creating a new code.	Flexible; the destination content or URL can be updated infinitely without changing the printed code.
Tracking and Analytics	Not trackable; provides zero visibility into user engagement or scan analytics.	Highly trackable; provides real-time analytics including scan counts, locations, dates, and device types.

Authentication Suitability	Suitable only for permanent, unchanging data.	Ideal for authentication due to support for short-lived, session-specific destinations, expiration dates, and scan limits.
Cost and Efficiency	Not cost-effective for changing data, as any update requires reprinting.	Highly cost-effective for evolving campaigns, saving manual effort and printing costs

Table 2.2 Compare table of dynamic and static QR code

2.6 Related technology

This section outlines the foundational Web3 technologies that power the decentralized ticketing platform:

I. NFTs or Non-Fungible Tokens refer to unique digitised assets that are stored permanently on a blockchain and prove ownership of the said digital asset. Unlike cryptocurrencies, NFTs are non-replaceable and have their own metadata. In this project, NFTs come equipped with smart contracts that enforce stringent rules regarding the secondary market which stops the problem of ticket fraud altogether.

II. Cryptocurrency is a distributed form of electronic exchange that utilizes the technology of distributed ledger to enable transactions without involving any traditional banking institutions. This allows for elimination of the need for centralized clearinghouses and the related high credit card fees and long waiting times for transactions. In a ticketing system deployed on the Web3 platform digital currency is required to pay for expenses associated with transaction processing and trading of tickets in an automated way.

III. A Web3 digital wallet refers to a decentralized software that enables users to manage their cryptocurrencies and NFTs while acting as the main entry point in their system. Unlike Web 2.O systems which rely on username and password systems deemed insecure, the wallet relies on a public-key private-key encryption mechanism thus ensuring its users' anonymity and security. When entering the event, the attendees prove that they own the ticket by signing transaction requests that protect their private key.

2.7 Existing Systems

For the project to create the groundwork for the system, the current system that is comparable to the project is crucial. The current system not only made it possible to comprehend the features, capabilities, and performance that were pertinent to the project field, but it also allowed for the comparison of various systems and the analysis of their strengths and weaknesses. This gave rise to the project's concept, which combined its benefits with enhancements to create a superior system.

2.5.1 TicketMaster

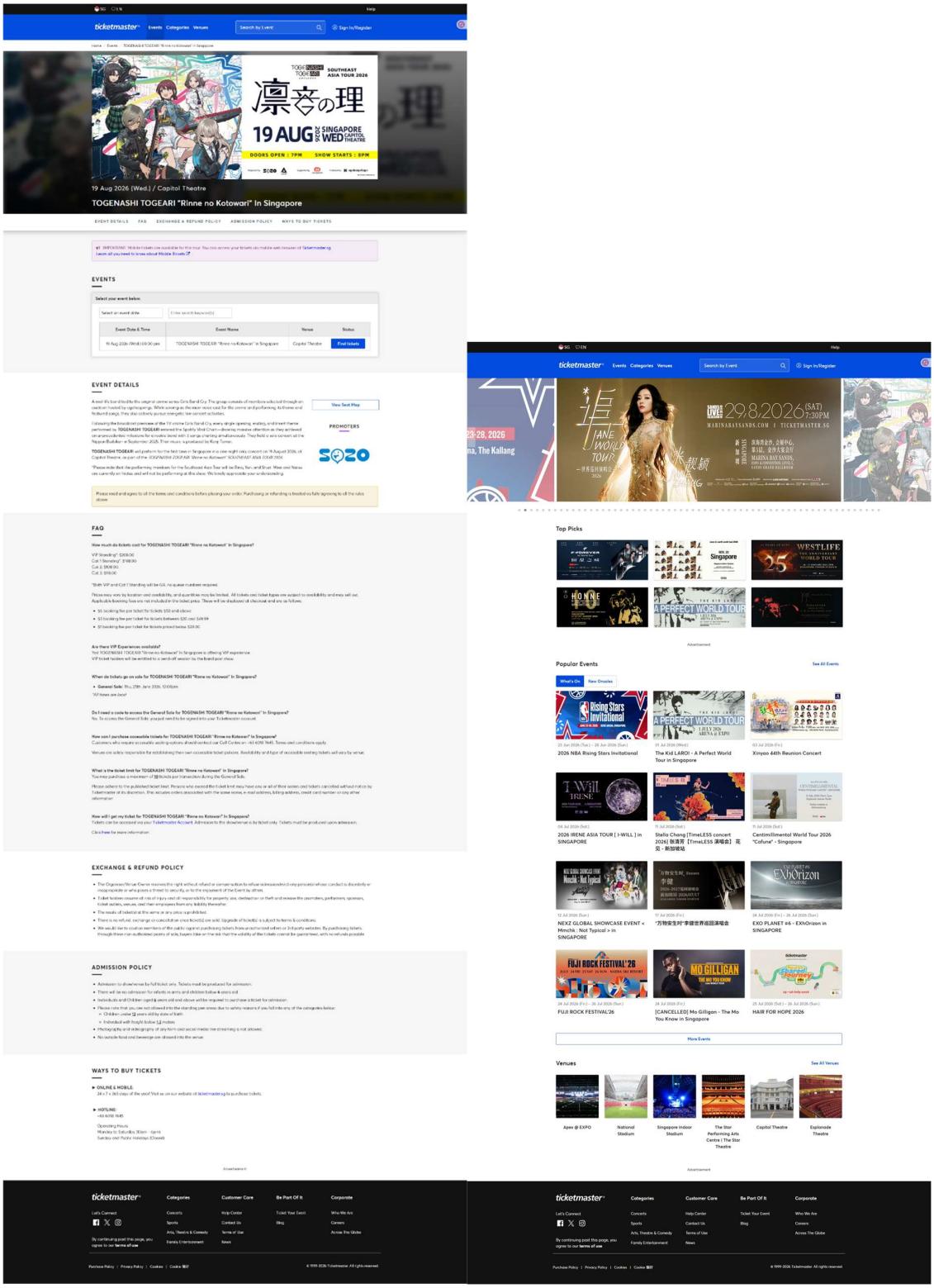


Figure 2.2 System screenshot of Ticket Master

The world's largest centralized electronic ticketing system and ticket distribution platform is Ticketmaster, which facilitates the connection between event venues across the globe and their

customers digitally. Since merging with Live Nation (Caesar J., 2026), Ticketmaster is now an example of vertical integration in the ticketing industry, as it has complete governance over the entire ticket life cycle, from primary ticket allocations to secondary resale management through its TradeDesk system. Ticketmaster's centralized relational databases provide a high level of security for ticketing activity, user data, and prices of available tickets during each event. Ticketmaster's real-time dynamic pricing algorithm is used to maximize the company's profits during periods where ticket demand is high by increasing prices as demand for tickets reaches certain predefined levels; this practice is used to direct the revenue from arbitrage on the secondary market back to the promoters and artists who originally created the demand for the tickets.

Ticketmaster utilizes its proprietary SafeTix technology to protect ticket gates. With SafeTix, Ticketmaster does not use static barcodes like other ticketing systems but instead generates dynamic QR codes via the client's mobile app (Caesar J., 2026). The QR code is encrypted and regenerates every few seconds with an algorithm that is defined by both randomness and time, which eliminates any duplicate visual representations, such as photographs taken on a cell phone or static copies of the ticket. Ticketmaster also restricts the purchasing of tickets in bulk, which is achieved through the use of the Verified Fan lottery system, which validates user or profile information against prior customer records in order to prevent automated internet purchases.

Although they have created a variety of defensive measures, Ticketmaster’s centralized and profit-oriented business model creates significant conflicts of interest. Investigations have found that the architecture of its platform allows brokers to use fake customer accounts to hoard inventory and create large-scale, secondary ticket loops. Because the company also indirectly profits from having multiple transaction fee loops also call as “triple-dipping” per single transaction of an asset (King A., 2025), there are no inherent incentives in the system to enforce complete compliance with the marketplace.

2.5.2 Ticket2U

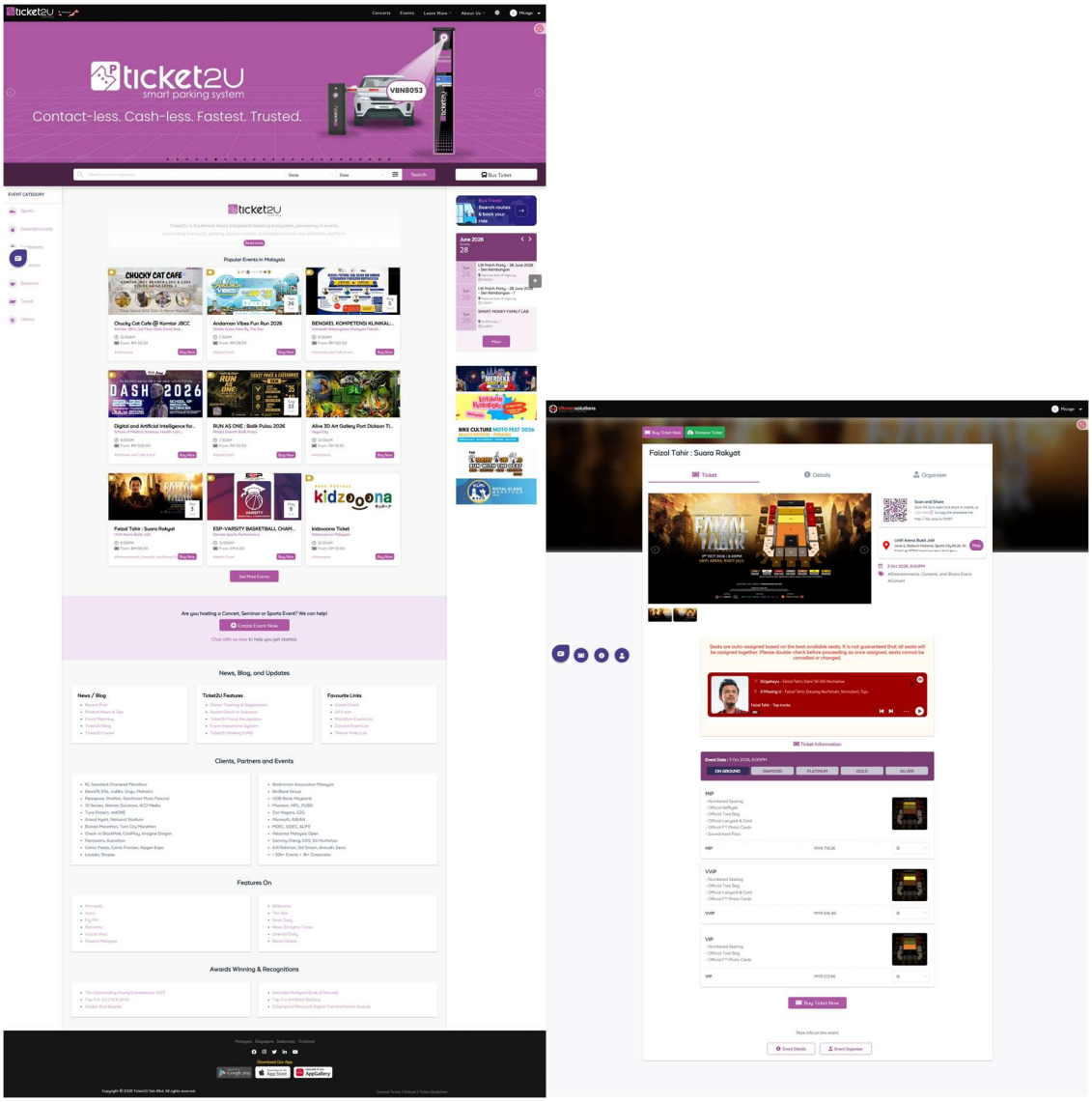


Figure 2.3 System screenshot of Ticket2U

Southeast Asia has Ticket2U as the leading SaaS ticketing and event management environment on the cloud. It has a concentrated regional customer base with a diverse array of markets such as live entertainment tours, marathons, theme parks and corporate exhibitions in Singapore and Malaysia. Its centralized cloud cluster operation provides an integrative experience for event planners through their event management platform, giving planners access to a variety of tools including interactive digital seating charts, real-time registration arrays for participants, automated lucky draw microservice functions and unique ability control logic. Event planners upload their event metrics through the platform and the centralized infrastructure natively handles the transaction processing for the organizers via a financial checkout and also allocates user inventory.

Ticket2U's high-volume, multi-faceted physical and digital verification of tickets is achieved through multiple endpoints. The core application layer relies on a proprietary API called Check-In Manager with a very low latency execution time of 0.01 seconds per e-ticket barcode or RFID asset and serve as an offline case sync when localized network connections fail to deliver the expected performance (Ticket2U, n.d.). Even more advanced access control features to the Ticket2U platform have been added through automated exhibition kiosks and turnstiles, which include facial recognition engines. The facial recognition engines cross-reference the user's facial biometric vector mapping at the gates with the platform's user(s) records in an array within 0.1 seconds to verify venue access authorization.

Ticket2U is heavily reliant on a centralized infrastructure to provide low-latency access management through physical entry. However, this reliance makes the Company vulnerable to many of the same risks that accompany centralized infrastructure, including database manipulation and identity fraud. Traditional email and messaging-based e-tickets provide little or no cryptographic protection against dynamic counterfeiting, thus creating opportunities for malicious double-selling. The lack of decentralized escrow enforcement creates opportunities for sellers and buyers to transact using peer-to-peer services on secondary markets but also makes it nearly impossible to monitor these transactions so exposing local participants to significant losses resulting from black-market scams.

2.5.3 GET Protocol

GET Protocol is a company that provides digital ticketing services based on blockchain technology. They have been around since 2016 and have sold more than 3 million tickets in over 120 countries (GET Protocol, 2023). GET Protocol has created a modular back-end that can be used by both independent ticket-generated providers and legacy centralized ticketing systems that wish to map their current inventories to the blockchain. All tickets that are sold through the GET Protocol system are automatically created as unique NFTs (Non-Fungible Tokens) using the Sophon and Base network, with all transactions being logged in real-time. Each completed transaction is recorded for all time in the public NFT Ticket Explorer and can be viewed there.

The verification process for accessing entry gates is based on the automated state execution of a smart contract in accordance with the GET Protocol framework. Your mobile user application has a secure, cryptographically secured key pair that associates your digital ticket asset directly with the authenticated state variables of your application instance. At the point of entrance into the venue, the scanning device communicates with the decentralized state machine to determine whether the currently active cryptographic signature is associated with the immutable owner address stored on-chain. If successful, this validates that your ticket has been checked-in and the smart contract will automatically transition the digital ticket to a collectible non-fungible token (NFT) which will be permanently stored in your decentralized Web3 wallet after the event.

While the GET Protocol effectively prevents scalping through rigid, programmable price controls and clear on-chain ownership boundaries, its dependence on a decentralised model creates additional trade-off challenges in the operation of GET Protocol. Using public mainnets to process state updates will require users to use native gas fees when accessing GET Protocol, meaning that the entire system may become very vulnerable to spikes in congestion on the public network. Moreover, due to the entire abstraction of using Web3 wallets for users without technical skills, the responsibility for privately storing private keys has been returned to GET Protocol's own centralised server solutions, creating a hybrid architecture that undermines principles of complete self-custody of funds with cryptography.

2.6 Comparison Matrix of Existing System

		TicketMaster	Ticket2U	GET Protocol
Account Management				
1	Register/Sign in by using Email	✓	✓	✓
2	Connect by using Web3 Wallet			✓
3	User Edit profile	✓	✓	✓
4	Pre-Registration Anti-Bot Screening	✓		
Verification Ticket Method				
1	QR Code (Dynamic/non-Dynamic)	✓	✓	✓
2	Barcode	✓	✓	
3	Face scanning		✓	
4	Online Smart Contract Signature Verification			✓
Features				
1	First-Hand purchase ticket	✓	✓	✓
2	Marketplace for selling ticket			✓
3	Organizer Create Event	✓	✓	✓
Additional Feature				
1	Automated Post-Event NFT Collectibles			✓
2	On-Chain Ticket Lifecycle Explorer			✓
3	Decentralized Event Financing (DeFi Loan)			✓
4	Integrated Onsite Kiosk & Badge Printing		✓	

Table 2.3 Compare table for existing system

2.6.1 Account Management

When comparing the entry management systems used by GET Protocol, Ticketmaster, and Ticket2U in terms of Web2/Web3 paradigm differences, it is evident that an entry layer has been established in each system for registering users as part of the onboarding experience. All three platforms include email registration at a base level that allows for the management of user registrations. The ability to create a cryptographic link between an external non-custodial Web3 wallet and a user account is provided only by GET Protocol through the use of tokenized user access at the account level. Also, the only platform with a formalized "Pre-Registration Anti-Bot Screening" process is Ticketmaster by virtue of their Verified Fan program (Caesar J., 2026), which uses machine learning to create a screening loop for identifying valid user accounts prior to high-demand ticket sales. In addition, for all platforms except GET Protocol, their systems use direct transactional queues for incoming user requests.

2.6.2 Verification Ticket Method

Diverse decisions made by software engineers about security, latency, and hardware dependency have been taken into account when implementing gate-access authentication of these three systems. All three systems utilize QR Codes either in a static or dynamic format for digital access validation purposes. While Ticketmaster and Ticket2U continue to use legacy static barcode media, which is still very vulnerable to cloning or being copied, Ticket2U uses localized biometric hardware systems that enable offline capable face scanning at the event gate turnstiles to meet the demand for fast regional access and scalability (Ticket2U, n.d.). In contrast, GET Protocol's online smart contract signature verification system provides a means to mitigate the vulnerabilities of physical hardware by authenticating ownership state through a decentralized public ledger signature as opposed to through the centralized server check of its structural ownership state.

2.6.3 Features

The transactions that make up the core of the centralized ticketing industry also identify some of the significant limitations of centralized versus decentralized ticketing. Each of the ticketing technologies being considered here supports the primary distribution networks in one-person limited ticket and also provide event creation dashboards for the organizer to create an event. The primary functions of these different systems are all limited to creating a second marketplace for selling tickets out of the official platform's boundaries without supporting a fully functional version of each. The last remaining centralized system for resale today was Ticketmaster; however, due to increasing amounts of legal intervention and continued issues with the systemic manipulation of the resale market through bots, Ticketmaster was forced to eliminate its TradeDesk secondary ticketing operation for concerts (Lind J.R., 2025). The only fully functional peer-to-peer regulated resale marketplace exists on GET Protocol's official Marketplace. Conversely, Ticket2U provides a fully restricted peer-to-peer resale marketplace that does not support any kind of native peer-to-peer resale market.

2.6.4 Additional Feature

The advanced service extensions illustrate how blockchain-based tokenization is more flexible architecturally compared to traditional centralized database models. For example, Ticket2U focuses purely on the logistics of physical events for enterprises and utilizes a fully integrated onsite hardware kiosk system that allows users to print badges and registration stickers immediately upon arrival at the event (Ticket2U, n.d.). On the other hand, all of the features that are available to Web3-based platforms will be limited to the smart contract ecosystem managed through GET Protocol. This includes but is not limited to: the automatic conversion of verified entry data into permanent, post-event NFT collectibles; complete structural transaction audits by utilizing an open, publicly accessible on-chain ticket lifecycle explorer; and the execution of decentralized event liquidity financing protocols that can provide upfront funding for event-related venues.

2.7 Framework



Solidity is an object-oriented, high-level programming language explicitly designed for developing self-executing smart contracts on the Ethereum Virtual Machine (EVM) (GeeksforGeeks, 2025). Serving as the core backend language of ChainTicket, Solidity is utilized to program the immutable logic governing ERC-721 ticket NFTs. It programmatically enforces the strict 10% prices in secondary market resale price ceiling and automates the instant distribution of split royalty fees back to event organizers. Its deterministic execution framework eliminates legacy database vulnerabilities, establishing an absolute, decentralized root of trust and eliminating corporate database manipulation.



Foundry is a blazing-fast, Rust-based development framework engineered for compiling, testing, and deploying EVM-compatible smart contracts (Weston G., 2023). In this project, Foundry serves as the main environment to optimize gas efficiency and execute robust security parameters before network deployment. It allows the developer to write comprehensive unit and fuzz tests directly in native Solidity, significantly boosting productivity. Foundry's advanced tracing, gas reporting, and localized EVM simulation capabilities ensure that anti-scalping protocols and dynamic validation states are rigorously debugged and mathematically secure prior to mainnet distribution.

2.8 Database



Firebase is a Google-built Backend-as-a-Service (BaaS) solution used to create and organize off-chain portions of a system's structured data (GeeksforGeeks, 2026). In the ChainTicket architecture, Firebase acts as a very large, NoSQL database that is customized to work as a complete user profile and event organizer profile repository. Furthermore, The ChainTicket system also uses the Firebase database to provide real-time analytic reporting of attendance metrics to event organizers as well as other important metadata about the event, including ticket release dates, current sales timelines, and exact venue layout specifications. By transferring high frequency, text-based querying and logistical updates from the ChainTicket system to low-latency streams of data in Firebase, the ChainTicket infrastructure delivers a more optimized Web2.5 user experience and significantly reduces the amount of gas consumed by its mainnet operations.



The Gnosis Chain serves as the Ethereum compatible, decentralized and distributed public ledger that will serve as the main immutable database for the entirety of the cryptographic ticket states (Gnosis, 2024). This records every aspect of absolute ownership of ticket minting (NFT's) as well as all transaction logs and lifecycle changes from when a ticket is minted until its permanent state is reached. By recording state changes on the Gnosis Chain's distributed network, we eliminate the ability to have a single failure point that comes from using a centralized approach. With an open and transparent data structure, we create an environment for security against institutional ticket overselling, unauthorized administrator interference with recorded transactions, and concealed black market distribution of tickets as well as complete public accountability.



Pinata is a Web3-based cloud hosting service created to supply developers with a powerful toolset for working with the IPFS (InterPlanetary File System) (Weston G., 2026). Storing large amounts of un-structured metadata for tickets (such as high-resolution images of the concert poster and complex JSON files for the seating chart for the venue) directly on-chain creates a lot of congestion and results in unsustainable gas prices. To address this engineering limitation, ChainTicket utilizes Pinata to offload its data to a permanent decentralized network that secures the data and produces a unique hash value (CID). Therefore, the Solidity smart contract only has to save the hash as the tokenURI, which allows for permanent storage of data that cannot be altered while only needing small amounts of infrastructure.

Chapter 3: Methodology

3.1 Introduction

The project's approach and process flow are covered in this chapter. The selection of the software process model, an introduction to the process model, and an explanation of each model step are provided in this chapter. This chapter outlines the FYP 1 timeframe, the hardware and software requirements for development, and the process framework. Each phase's goals and flow will be described in the process framework.

3.2 Agile Methodology

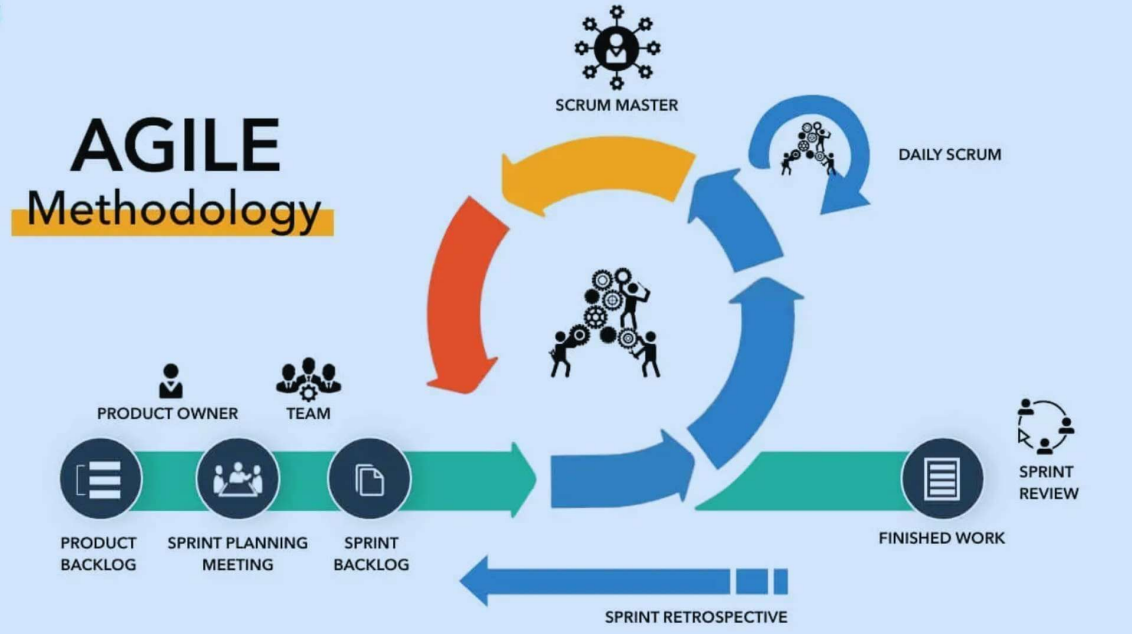


Figure 3 Agile Mothodology

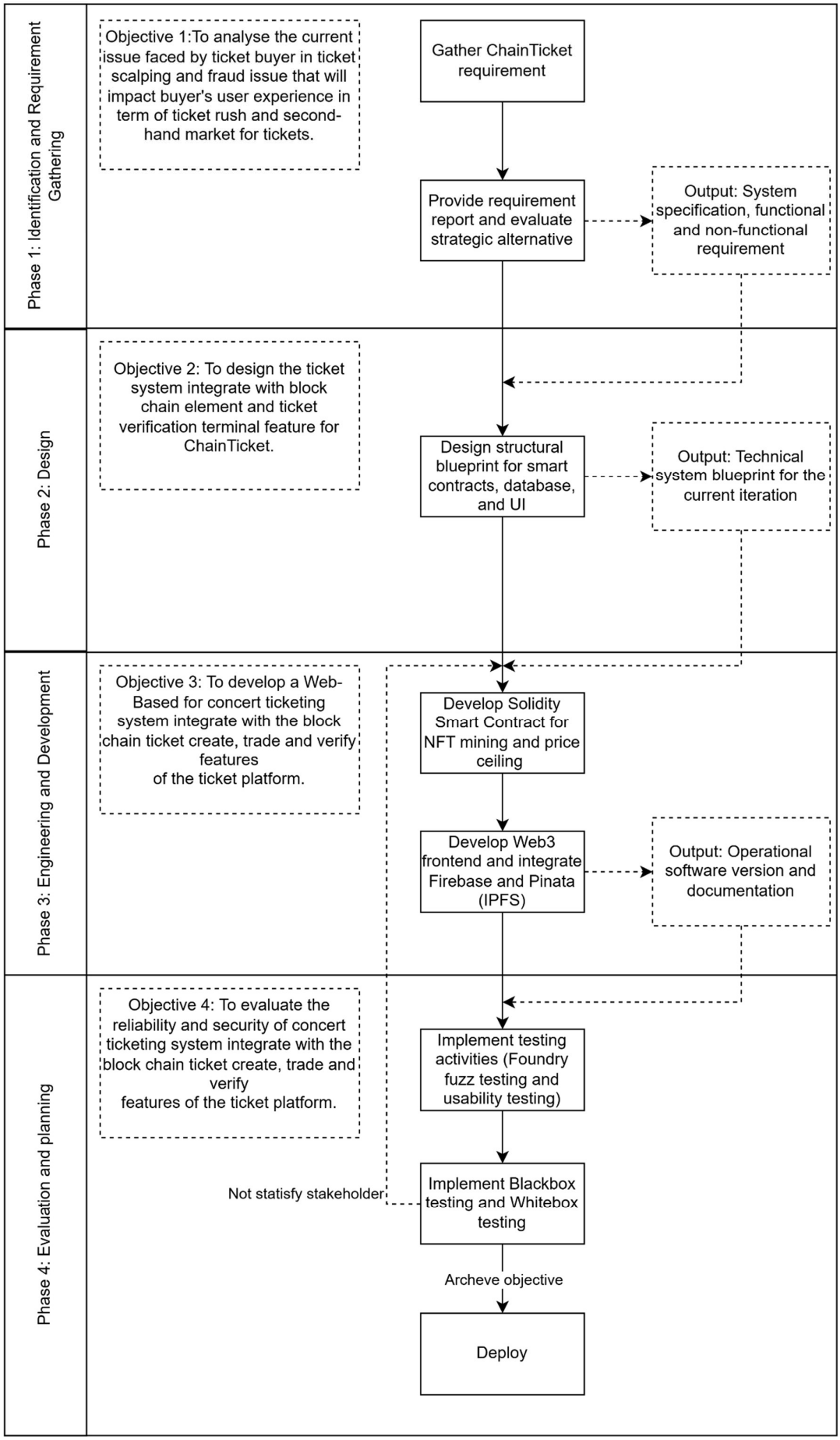
Agile Methodology with Scrum framework serves as the principle methodology in this project. This technique perfectly fits unpredictable projects, which need constant adaptability, iteration, and flexibility. In contrast to a more traditional mode, Agile method is not linear; it is a collaborative method with short intervals of development called sprints. After being led by the Scrum Master during the development process, the team collaborates tightly with the Product Owner in order to create maximum value continuously. Thus, thanks to the presented cycle it becomes much easier to define the technical problems or security issues at the beginning of every sprint.

The operational process adheres to a standard Scrum life cycle pattern. The first step involves creating a backlog whereby the inputs are obtained from the Product Owner, who works with stakeholders to come up with the Product Backlog, which is a comprehensive list of all the functionalities required for the system. After a Sprint Planning Meeting, the top priority items are moved

to the Sprint Backlog whereby goals for the current iteration are defined. The second stage is the design and development phase whereby the development team focuses on programming and creating smart contracts according to the current objectives of the Sprint. During this active stage, the team holds a Daily Scrum meeting to check on the progress made daily as well as address and resolve any issues in the development process. The third stage is the delivery phase whereby the team provides Finished Work, which represents a completely functional product. The final step is the review and retrospective phase whereby the team conducts a Sprint Review to show the finished work to the Product Owner and collect valuable feedback from the customer. After the Sprint Review, the team conducts a Sprint Retrospective in order to analyse their internal development process and make plans on how they can improve their operations in the next Sprint cycle.

Choosing the Agile methodology, specifically implementing the Sprint framework, is warranted considering the nature of the development of the decentralized Web2.5 ticketing platform. Unlike conventional linear methodologies, Agile offers essential flexibility and adaptability needed to address the uncertainties arising in the course of blockchain integration as well as smart contract implementation. With the Agile methodology, the project development life cycle is broken down into small iterations called sprints, thus making it possible to build, test and adjust such core features as NFT minting, QR code verification, and secondary market pricing protocols continuously. Such an iterative approach allows spotting and addressing any technical challenges or security threats from the very beginning of the development. What is more, the collaborative nature of Agile methodology allows the development team and stakeholders to exchange feedback constantly during the Sprint Review phase.

3.3 Workflow overview



Phase 1: Identification and Requirement Gathering focuses on analysing user challenges like ticket scalping and fraud. It involves gathering system requirements and evaluating strategic alternatives to produce a final system specification detailing functional and non-functional requirements.

Phase 2: Design translates these specifications into a structural blueprint. This phase designs the architecture for the smart contracts, databases, and user interfaces, culminating in a technical blueprint for the current development iteration.

Phase 3: Engineering and Development brings the design to life by programming Solidity smart contracts for NFT minting and enforcing secondary market price ceilings. Concurrently, the Web3 frontend is built and integrated with Firebase and Pinata (IPFS), resulting in an operational software version and documentation.

Phase 4: Evaluation and Planning ensures system reliability and security through rigorous testing, including Foundry fuzz testing, usability testing, and both black-box and white-box testing. If the outcomes do not satisfy stakeholders, the process loops back to the development phase for revisions. Once all objectives are successfully achieved, the final system proceeds to deployment.

3.3.1 Sprint 1

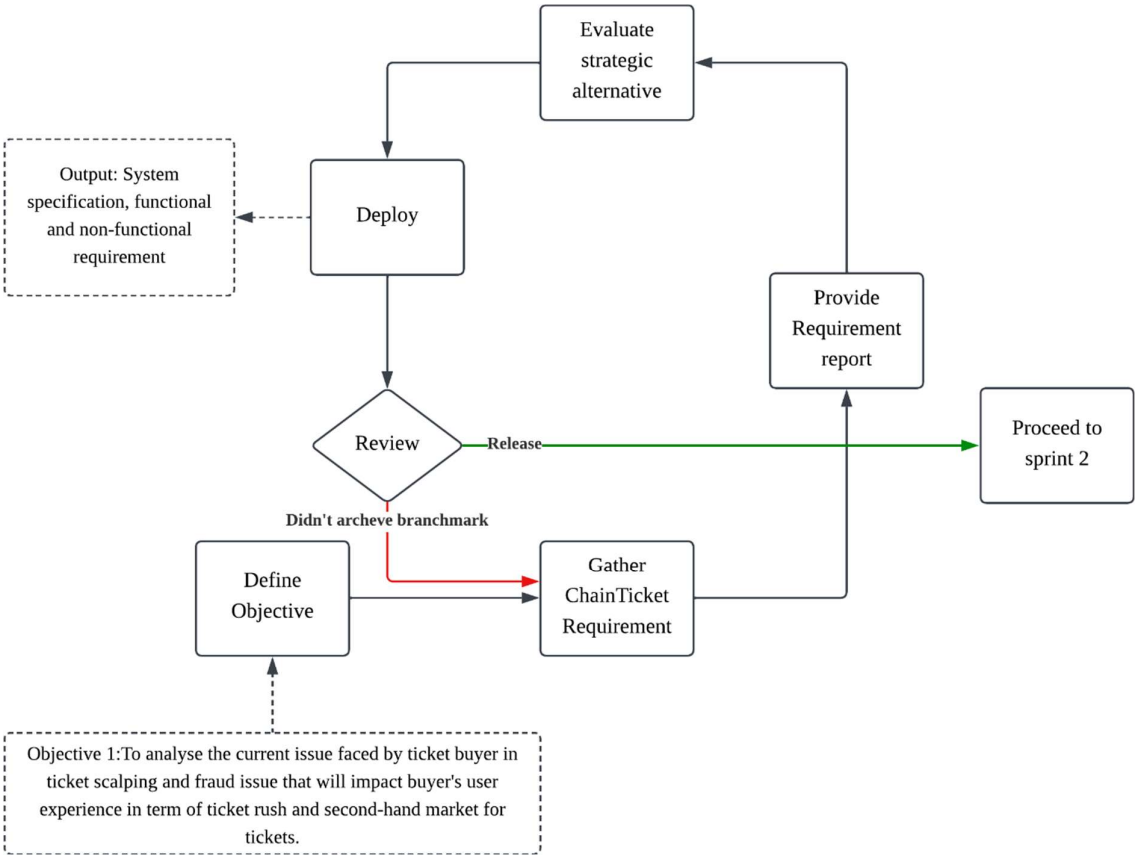


Figure 3.1 Agile sprint 1

In the sprint 1, which focus on identification and requirement gathering, the primary objective is “To analyse the current issue faced by ticket buyer in ticket scalping and fraud issue that will impact buyer's user experience in term of ticket rush and second-hand market for tickets”. This initial phase will focus on understanding the existing challenge and define the new feature and system that need to be achieve in this study. In this phase, the task is to identify the requirement and specification by gathering the ChainTicket requirement and features by conducting the analysing existing system. This phase will gather requirement of event ticket system and the integrate with the block chain. During this phase, the challenge that face by the project will be found out and analyse to ensure the entire project succession. After this, the gathered information will be compiled into a comprehensive report to outline the system functional and non-functional requirement.

3.3.2 Sprint 2

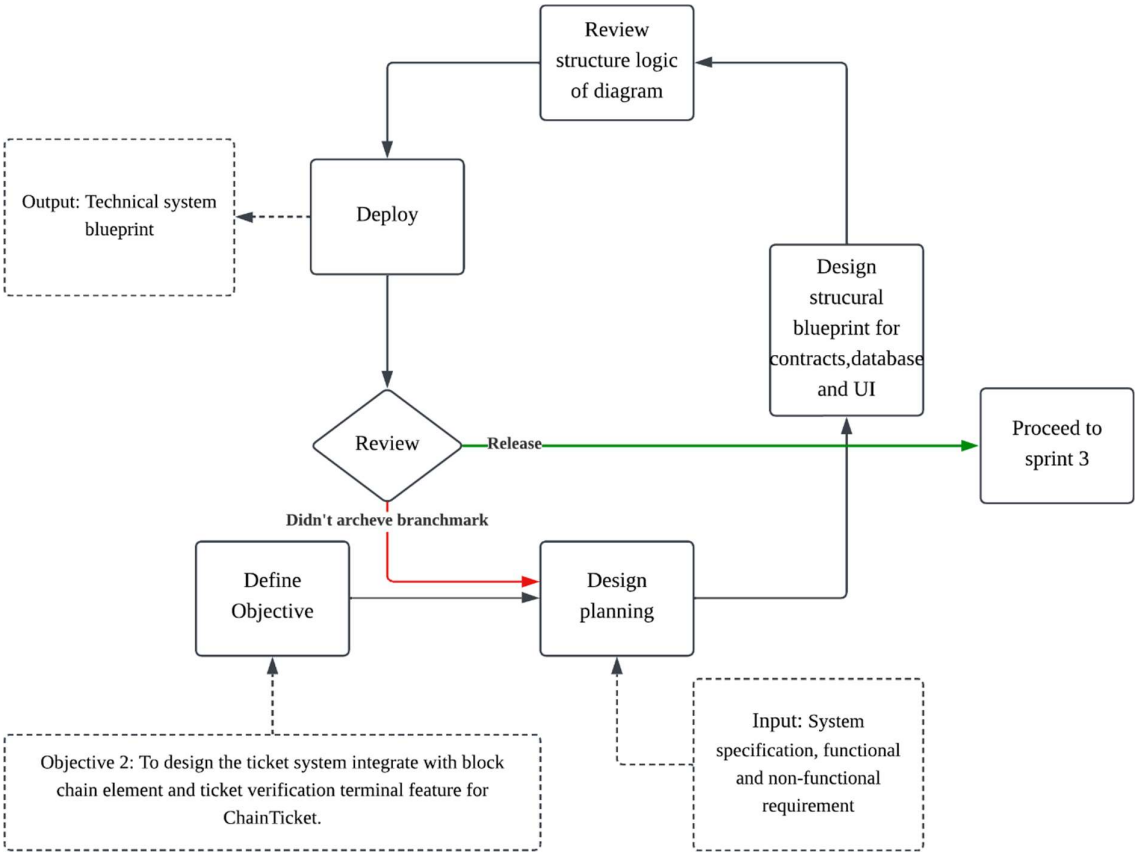


Figure 3.2 Agile sprint 2

In sprint 2, which focuses on design, the primary objective is "To design the ticket system integrated with blockchain elements and ticket verification terminal features for ChainTicket." The task in this stage is going to consist in defining the whole infrastructure of the above-mentioned ticketing platform according to the received demands. The main objective of this stage is the design of the smart contract protocol, ticket confirmation processes and the interface for the users according the requirement list form sprint 1. It will also include the idea of how the NFTs have to be created on the blockchain and how to properly confirm the access rights through some kind of special confirming terminal. The design of the interface will also be important to fulfil so that the work with the platform would be user-friendly and efficient. As a result of the design stage, the infrastructure of the project will be elaborated and well-prepared.

3.3.3 Sprint 3

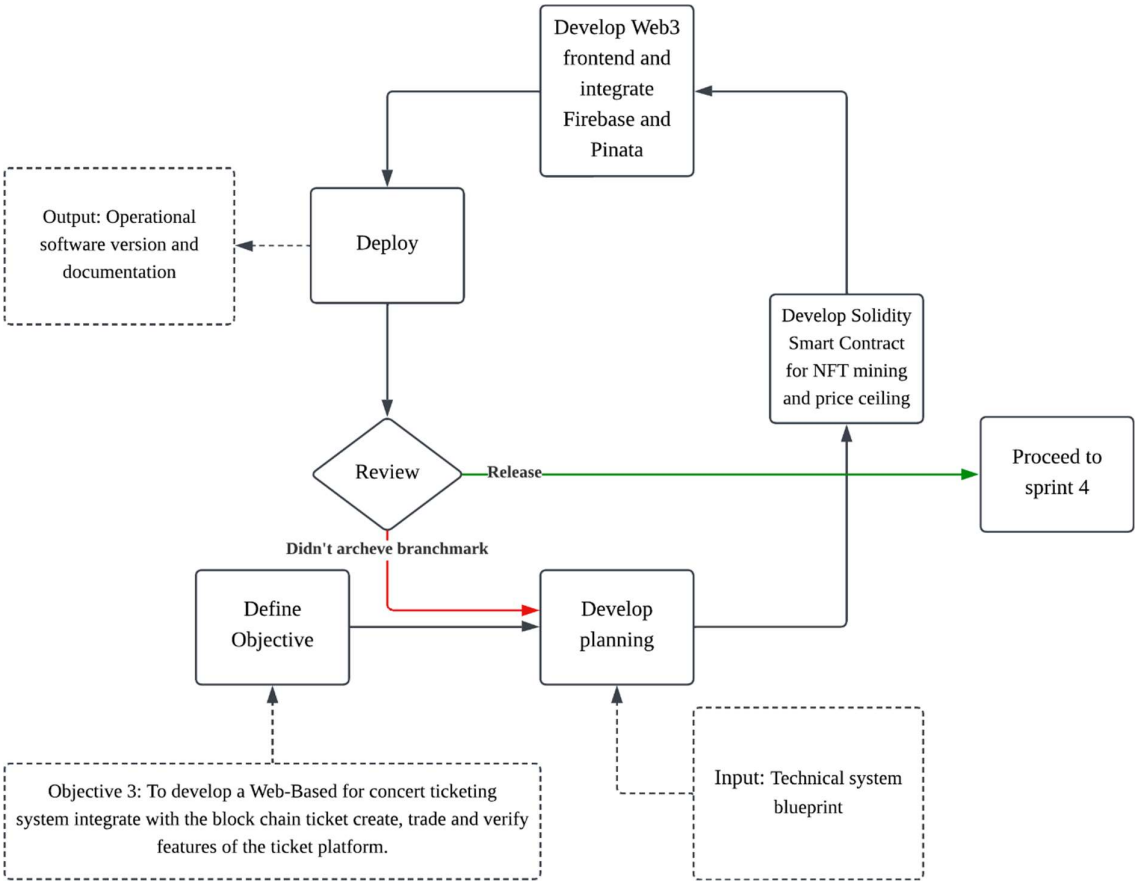


Figure 3.3 Agile sprint 3

In sprint 3, which focuses on engineering and development, the primary objective is "To develop a Web-Based concert ticketing system integrated with blockchain ticket creation, trading, and verification features for the ticket platform." This phase will transform the created designs into reality through the build-up of Web-Based platform based on usability and safety aspects of Web3 interactions according to the design documentation and technical system blueprint. At this stage, the basic logic and mechanism of the workings of the system will be programmed along with smart contracts coded in Solidity for NFT minting and price limitations for the secondary market. As a result of this phase, the basic working software will be generated, together with documentation about it for the next process step. The next process will be the creation of the front end system on the basis of previously created smart contracts.

3.3.4 Sprint 4

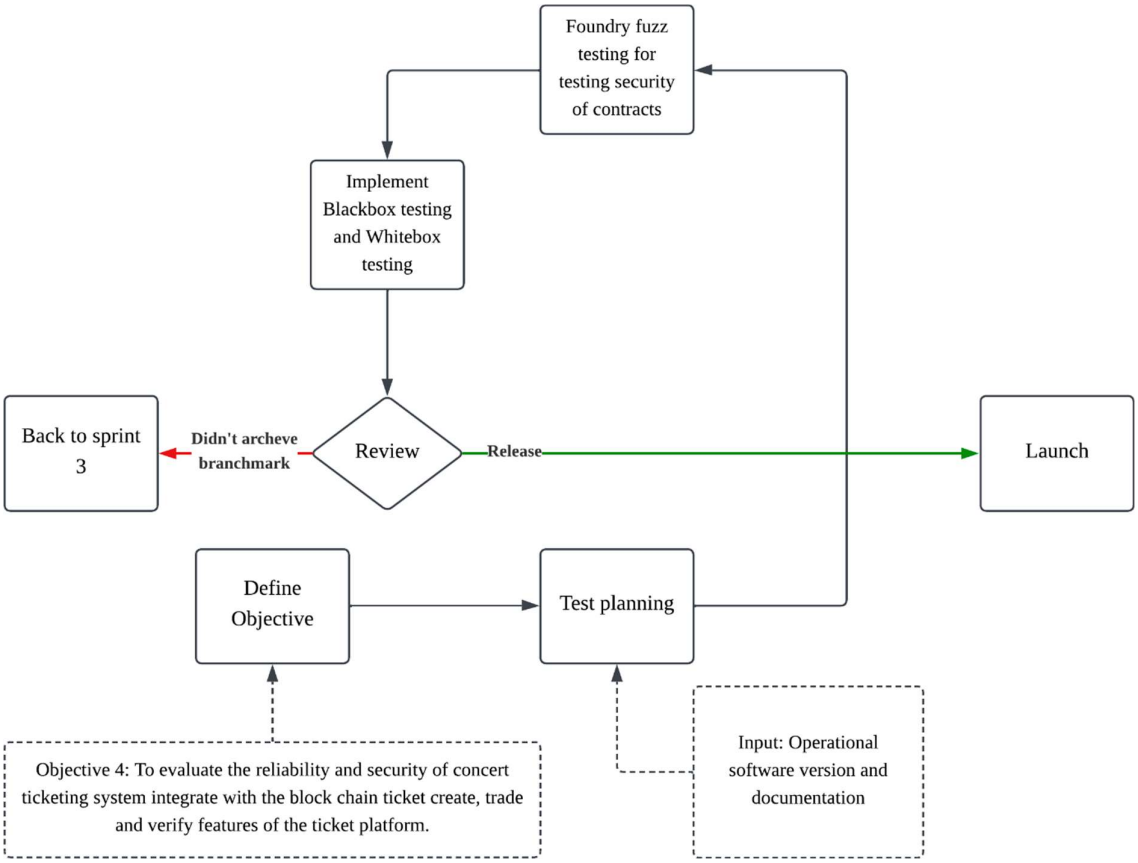


Figure 3.4 Agile sprint 4

In sprint 4, which focuses on Evaluation and Testing, the primary objective is "To evaluate the reliability and security of the concert ticketing system integrated with the blockchain ticket creation, trading, and verification features of the ticket platform." This phase involves ensuring that all components function correctly through thorough testing and verifying their readiness for the platform. The testing phase will comprise the identification of smart contract bugs and usability issues, among others. The integration will implement the different Web-app and frontend modules coded and unit tested. The Blackbox testing and Whitebox testing will be conduct for all stakeholder for verify the usability and the function of system, if the testing result does not satisfy the quality requirement the process will backword to sprint 3 to engineering and develop again. After pass all the testing and verification of the NFT ticketing and verification function, the software iteration will be launched for users.

3.4 Gantt Chart

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Project proposal														
Chapter 1: Introduction														
Chapter 2: Literature Review														
Chapter 3: Methodology														
Chapter 4: Design														
Chapter 5: Implementation														
Chapter 6: Conclusion														
Submit Full Report														
Presentation														
Working status										Review Status				

Table 3 Gantt Chart

The Gantt chart for the FYP 1 execution is shown in the table above. FYP 1 began on May 28, 2025, and ran for about 14 weeks, ending on August 25, 2025. Each of the chapter's tasks takes around a week, but because they are more complicated and require more labor, the design and initial implementation will take two to three weeks. Following the completion of the work chapter, the review phase will be put into place to guarantee that the project details and documentation are accurate, comprehensive, and consistent.

The FYP 1 process consists of 9 phases that must be finished. The first phases is the project proposal. The project title and concept will be submitted to the FYP moderator in the first week. In week two, the first chapter will introduce the system and provide a quick review of the background study. In week three, the second chapter will cover the background information and give a general overview. The chapter two procedure will run concurrently with the chapter one modification. The project's methodology and process flow will be covered in Week 4, Chapter 3. The chapter three procedure will run concurrently with the chapter two amendment. The system diagram, which serves as the development's guide, will be shown in Week 5, Chapter 4. The chapter four procedure will be parallel to the chapter three modification. In order to provide a visual depiction of the complete system, a low fidelity prototype of the system will be created in Week 7, Chapter 5. The chapter five procedure will be parallel to the chapter four modification. In week twelve, the thesis's conclusion will be summed

up based on the system as a whole, highlighting its shortcomings and suggestions for future development. In week thirteen, following the finalization of the adjustment with the supervisor, the submission was required so that the supervisor may evaluate and comment. The presentation will be completed in week 14 on the ongoing day.

Chapter 4: Design

4.1 Introduction

In this chapter, the design diagrams of the system shall be presented, which will serve as a guide for the development of the system. The different kinds of design diagrams help in understanding the behavior and the structural architecture of the system before the programming and implementation come into the picture. The chapter will contain various UML diagrams such as the Use Case Diagram, Class Diagram, Sequence Diagram, and Activity Diagram.

The Use Case diagram captures the usability and the requirements of the system and shows how the actors of the system interact and communicate. The Class diagram discusses the structure of the system and presents the available objects, their attributes, functions, and relationships between different classes. The Sequence diagram provides a view of the system's dynamics and represents the sequence in which messages and interactions between actors or objects take place. Finally, the Activity diagram shows how the different tasks are executed in the system in terms of activities and conditions required for the decision-making process in various situations.

4.2 Use Case Diagram

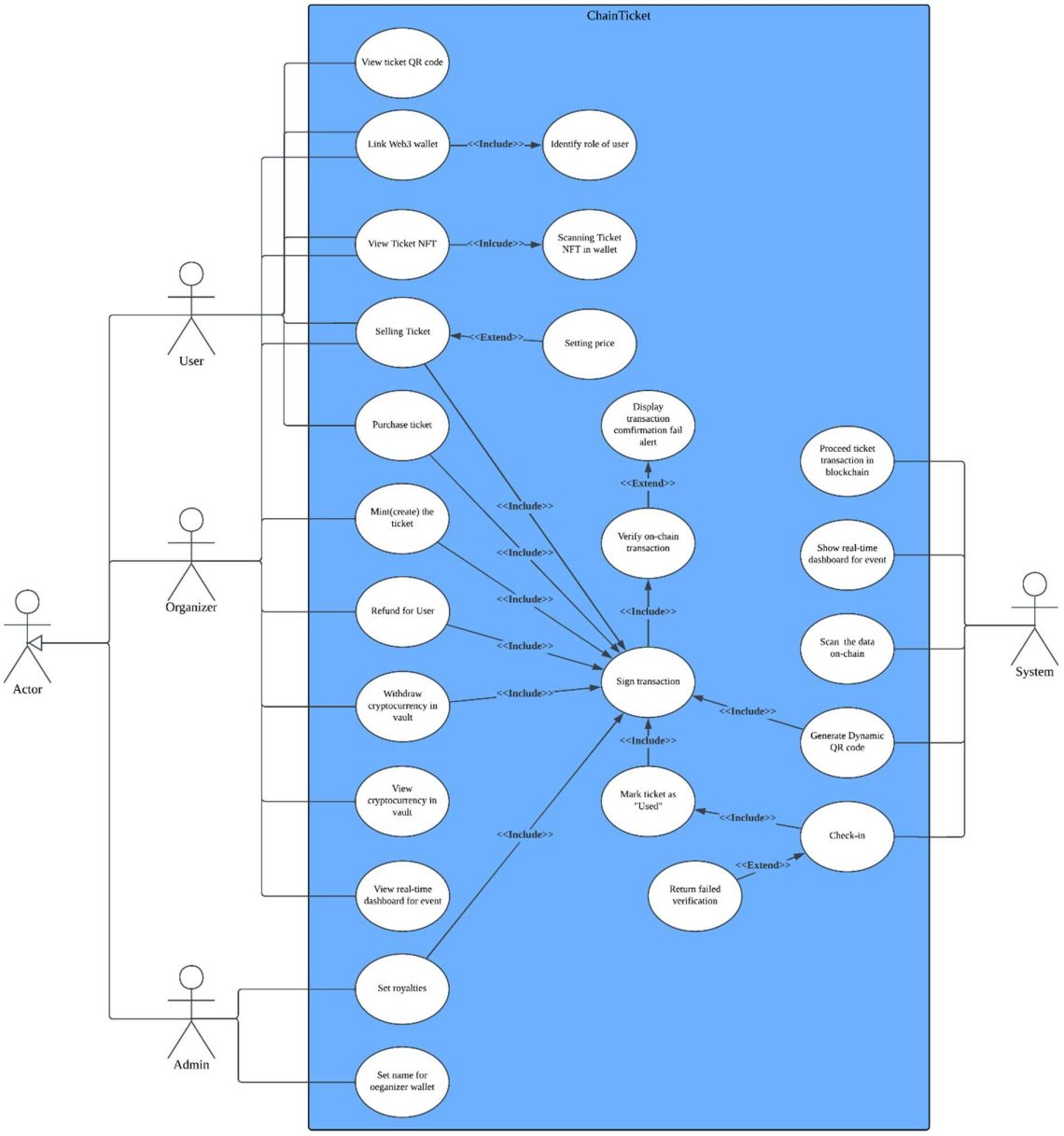


Figure 4.1 Use case diagram

4.2.1 [CA001] Purchase Ticket

[Use Case ID] Name	[CA001] Purchase Ticket
Description	Client/User can purchase ticket form marketplace
Actor	User

Pre-condition	1. The user connected the wallet. 2. The wallet has enough gas fee and amount. 3. The user trade in marketplace.
Post-condition	1. The NFT transferred to user wallet.
Main Success Path	
Actor Action	**System Response**
1. The user transfers the fund to vault	1. The system shows the purchased NFT ticket 2. The system mapping the NFT

4.2.2 [CA002] Selling Ticket

[Use Case ID] Name	[CA002] Selling Ticket
Description	User and organizer can list their owned ticket for sale on the secondary market.
Actor	User, Organizer
Pre-condition	1. The page connected the Web3 wallet. 2. The wallet has enough gas fee. 3. The user owns a valid Ticket NFT.
Post-condition	1. The ticket is successfully listed for sale on the marketplace.
Main Success Path	
Actor Action	**System Response**
1. The user selects the "Selling Ticket" option and executes the "Setting price" action. 2. The user proceeds to "Sign transaction" via their Web3 wallet.	1. The system executes "Verify on-chain transaction". 2. The system lists the ticket with the specified price.

4.2.3 [CA003] Mint Ticket

[Use Case ID] Name	[CA003] Mint Ticket
Description	Organizer can mint (create) the ticket for an upcoming event.
Actor	Organizer
Pre-condition	1. The organizer connected the Web3 wallet. 2. The wallet has enough gas fee and amount.
Post-condition	1. The Ticket NFTs are successfully minted and recorded on the blockchain.
Main Success Path	
Actor Action	**System Response**
1. The organizer initiates the "Mint(create) the ticket" process. 2. The organizer proceeds to "Sign transaction" to approve the minting.	1. The system executes "Verify on-chain transaction". 2. The system executes "Proceed ticket transaction in blockchain".

4.2.4 [CA004] Check-In event

[Use Case ID] Name	[CA004] Check-In
Description	The system verifies the ticket validity for event entry and updates its status.
Actor	System
Pre-condition	1. A Dynamic QR code has been generated and presented. 2. The ticket status is currently unused.
Post-condition	1. The ticket is marked as used to prevent duplicate entry.

Main Success Path	
Actor Action	**System Response**
1. (Triggered via scanner) The dynamic QR code data is submitted for the "Check-in" process.	1. The system processes the "Check-in" request. 2. The system executes "Mark ticket as "Used"" to permanently update the ticket's state.

4.2.5 [CA005] View Real-time Dashboard

[Use Case ID] Name	[CA005] View Real-time Dashboard
Description	Organizer can view the live data and check-in status of an ongoing event.
Actor	Organizer
Pre-condition	1. The organizer connected the Web3 wallet. 2. The event has been created and has associated ticket data.
Post-condition	1. The real-time statistics and dashboard for the event are successfully displayed to the organizer.
Main Success Path	
Actor Action	**System Response**
1. The organizer selects to view the real-time dashboard for a specific event.	1. The system executes "Show real-time dashboard for event" to retrieve and display live on-chain and off-chain data.

4.2.6 [CA006] Refund the cryptocurrency

[Use Case ID] Name	[CA006] Refund the cryptocurrency

Description	Organizer initiates a process to refund the cryptocurrency to users for unsold tickets or cancelled events.
Actor	Organizer
Pre-condition	1. The organizer connected the Web3 wallet. 2. The wallet has enough gas fee and amount to process the refund.
Post-condition	1. The smart contract successfully executes the refund, transferring cryptocurrency back to the users' wallets.
Main Success Path	
Actor Action	**System Response**
1. The organizer initiates the "Refund for User" action. 2. The organizer proceeds to "Sign transaction" via their Web3 wallet.	1. The system executes "Verify on-chain transaction". 2. The system executes "Proceed ticket transaction in blockchain" to process the refund.

4.2.6 [CA007] Withdraw cryptocurrency from vault

[Use Case ID] Name	[CA007] Withdraw cryptocurrency from vault
Description	Organizer withdraws the earned cryptocurrency (ticket sales and royalties) from the smart contract vault to their own wallet.
Actor	Organizer
Pre-condition	1. The organizer connected the Web3 wallet. 2. The wallet has enough gas fee to execute the withdrawal.

	3. The vault has an available cryptocurrency balance.
Post-condition	1. The cryptocurrency is successfully transferred from the smart contract vault to the Organizer's Web3 wallet.
Main Success Path	
Actor Action	**System Response**
1. The organizer initiates the "Withdraw cryptocurrency in vault" action. 2. The organizer proceeds to "Sign transaction" via their Web3 wallet.	1. The system executes "Verify on-chain transaction". 2. The system executes "Proceed ticket transaction in blockchain" to transfer the funds to the organizer.

4.3 Sequence Diagram

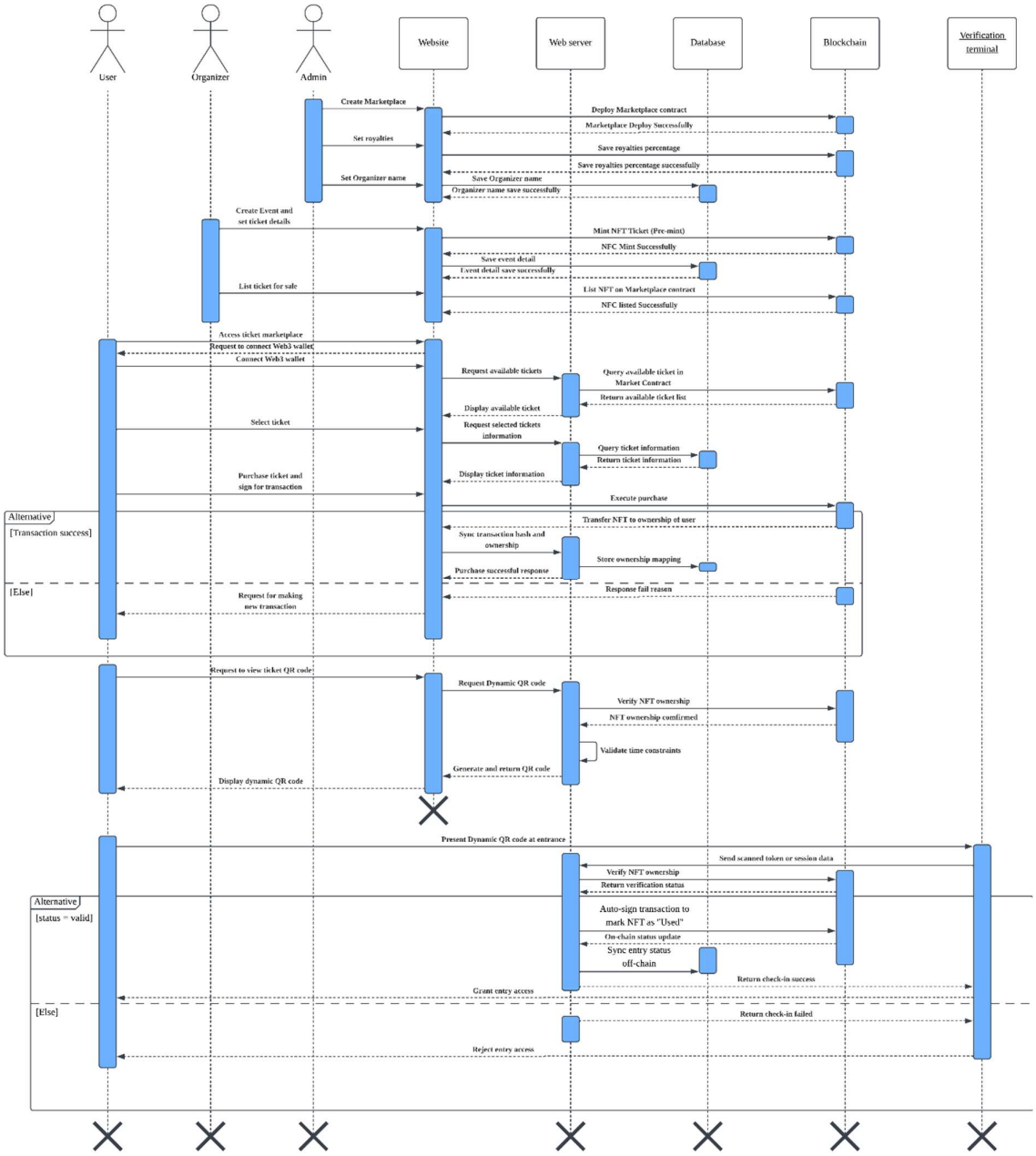


Figure 4.2 Sequence Diagram

4.3.1 System setup and configuration

The procedure starts with an Admin making modifications to the global system settings through the Website. The admin lays the foundation for the marketplace, which results in the deployment of the Marketplace contract on the Blockchain. The deployment is followed by the action of the admin who specifies the royalty percentage, and the data is recorded directly on the Blockchain. Finally, the name of the organizer is recorded by the admin into the off-chain Database through the Website.

4.3.2 Event creation and ticketing

The Organizer utilizes the Website to initiate a new event and configure the ticket specifications. This procedure prompts the Website to initiate the minting of NFT tickets on the Blockchain directly. When the minting is completed successfully, the event details out of Blockchain for example, venue and time will be stored in the Database. The Organizer then sells the minted tickets, which allows the Website to change the status of the Marketplace agreement on the Blockchain.

4.3.3 Ticket purchase flow

By using the Website, the User is directed to connect to their Web3 wallet and once the wallet is connected, the User requests to see available tickets. The server then queries the Market Contract on the Blockchain for available tickets and pulls in some ticket information from the Database.

The User then selects the ticket and makes the purchase by signing for the transaction. The purchase is made on the Blockchain, prompting the formation of an Alternative (ALT) block as per the result of the transaction:

- [Transaction success]: The NFT gets transferred to the User through the Blockchain. The Web server records this transaction and the ownership in the Database.
- [Else / Failure]: The Blockchain returns a reason for the failure and the Website prompts the User to start the transaction again.

4.3.4 Dynamic QR Code generation

When a user intends to participate in an event, they ask the website to show them the QR code for their ticket. This request is sent to the web server, which consults the blockchain to validate the ownership of the user's NFT. After having verified ownership, the web server checks for time constraints such like ensures that it is not too far from the time of the event, generates a dynamic QR code, and sends it back to the website to present to the user.

4.3.5 Check-In and verification

At the entrance to the event location, the User presents the Dynamic QR code at the verification terminal. As the terminal scans the token or session information, it sends it to the Web server, which polls the Blockchain for current NFT ownership verification. The Blockchain returns verification results which can result in a final Alternative block:

- [status = valid]: The Web server automatically signs the transaction of setting the NFT status to "Used" on the Blockchain. Once the on-chain status is updated, the Web server will synchronize this status offline with the Database. It then sends the successful check-in message to the terminal which means that the User is granted an entry.

- [Else / Invalid]: The Web server sends the check-in failure message to the terminal, and the User is denied entry.

4.4 Class Diagram

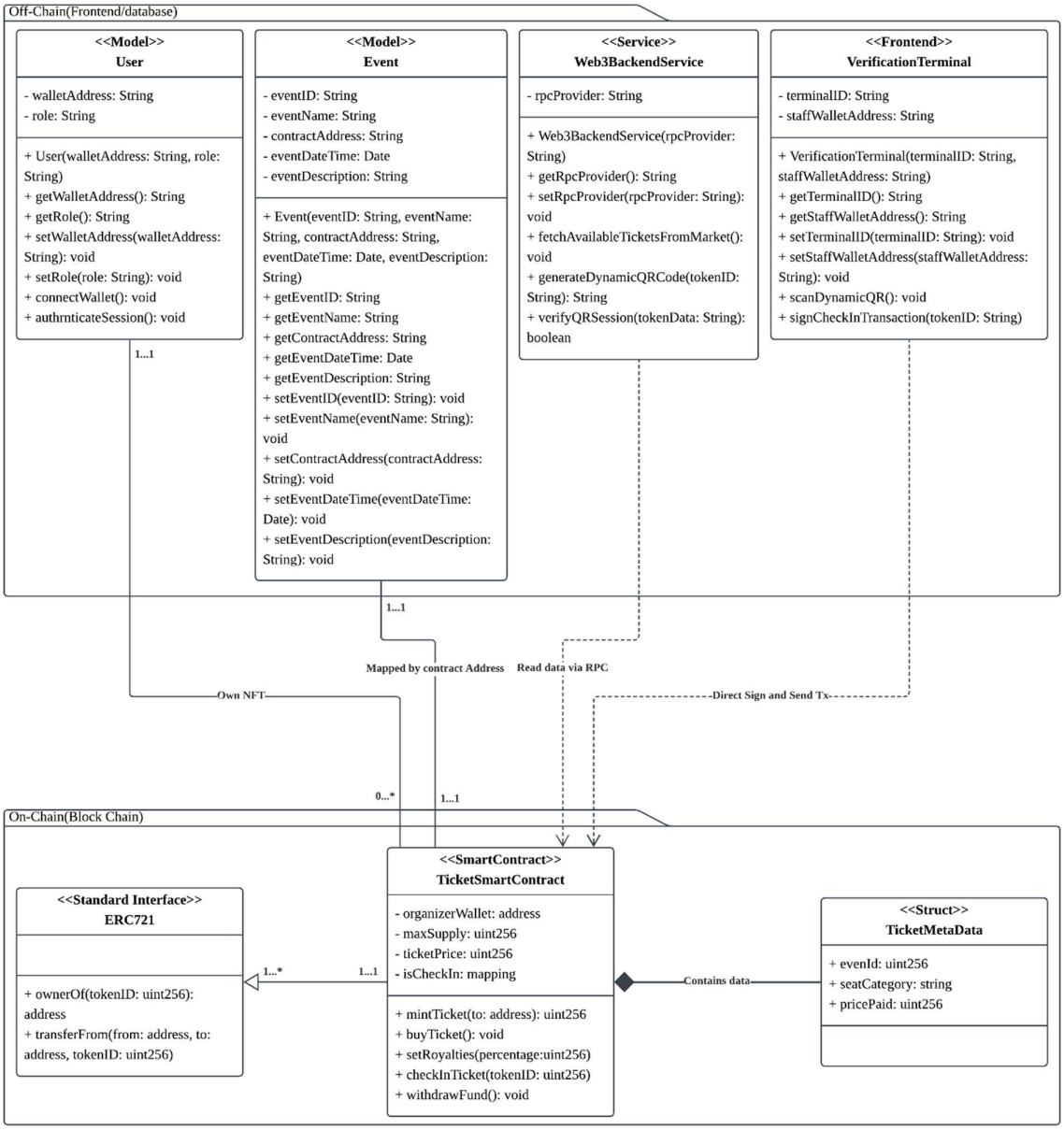


Figure 4.3 Class diagram

The class diagram illustrated in Figure 4.3 comprises the classes, attributes, and relationships of the system. The architecture of the system has two major environments: Off-Chain (Frontend/database) and On-Chain (BlockChain). The class diagram consists of several important classes, such as User, Event, Web3BackendService, VerificationTerminal, TicketSmartContract, and TicketMetaData. A User can own several Ticket NFTs. In contrast, an Event is uniquely associated with only one TicketSmartContract.

As for the User class, it has attributes like walletAddress and role. In addition, the class uses the methods connectWallet() and authenticateSession() for the verification of the user's identity. In order to buy or hold the NFT tickets, the user collects information about the Web3 identity.In the case of Event class, it involves all the off-chain data and consists of the attributes eventID, eventName, contractAddress, eventDateTime, and eventDescription.

According to the service and front-end classes, the Web3BackendService is a mediator that obtains data from the RPC, thus it has the rpcProvider property and calls methods such as fetchAvailableTicketsFromMarket(), generateDynamicQRCode(), and verifyQRSession(). The responsibility of the VerificationTerminal class is to oversee all staff-associated operations, therefore it possesses such properties as terminalID and staffWalletAddress, plus it applies scanDynamicQR(), signCheckInTransaction() in order to sign the check-in transaction and send it for processing to Blockchain.

With respect to On-Chain aspect, the TicketSmartContract is an important class that implements the pattern ERC721, thus it can utilize such standard NFT methods as ownerOf() and transferFrom() The TicketSmartContract has attributes such as organizerWallet, maxSupply, ticketPrice, and the isCheckIn mapping. It employs the following core methods of Blockchain: mintTicket(), buyTicket(), setRoyalties(), and checkInTicket The TicketMetaData (eventId, seatCategory, pricePaid) is closely tied together with the TicketSmartContract because in case this smart contract does not exist or its lifecycle is over, the ticket meta data is supposed to become non-existent as well.

4.5 Activity Diagram

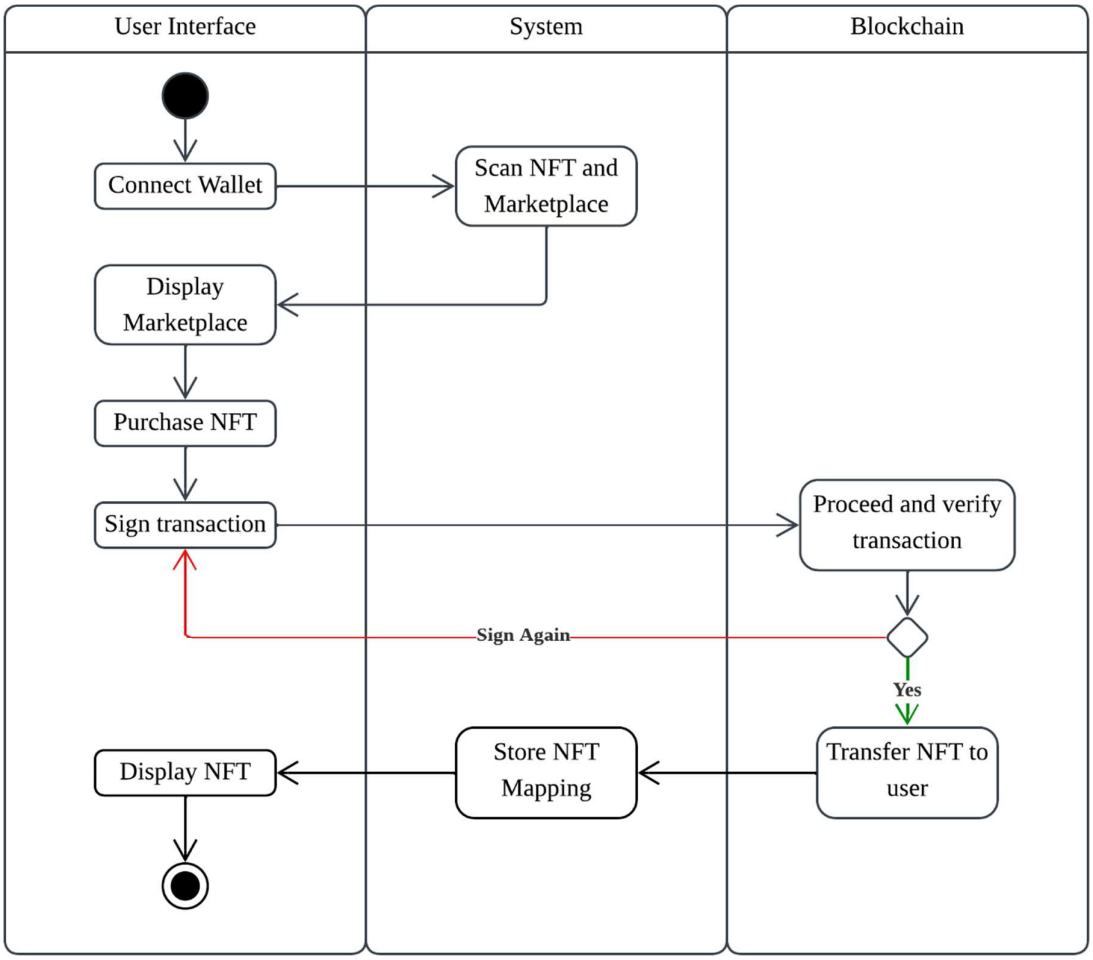


Figure 4.4 Activities diagram about purchase ticket

The purchase activity related to NFTs is illustrated in Figure 4.4. After the user connects the wallet using the user interface, the system scans the NFT and the marketplace. After that, the user interface shows the marketplace. When the user chooses to make the purchase of an NFT, they will have to authorize the transaction via signing it. The blockchain will continue to confirm the transaction. If the confirmation fails, the user will be required to re-sign the transaction again. In case the confirmation is successful, the blockchain will deliver the NFT to the user. The system is going to keep the record of the NFTs and the user interface is going to show the NFT.

4.6 Activity Diagram

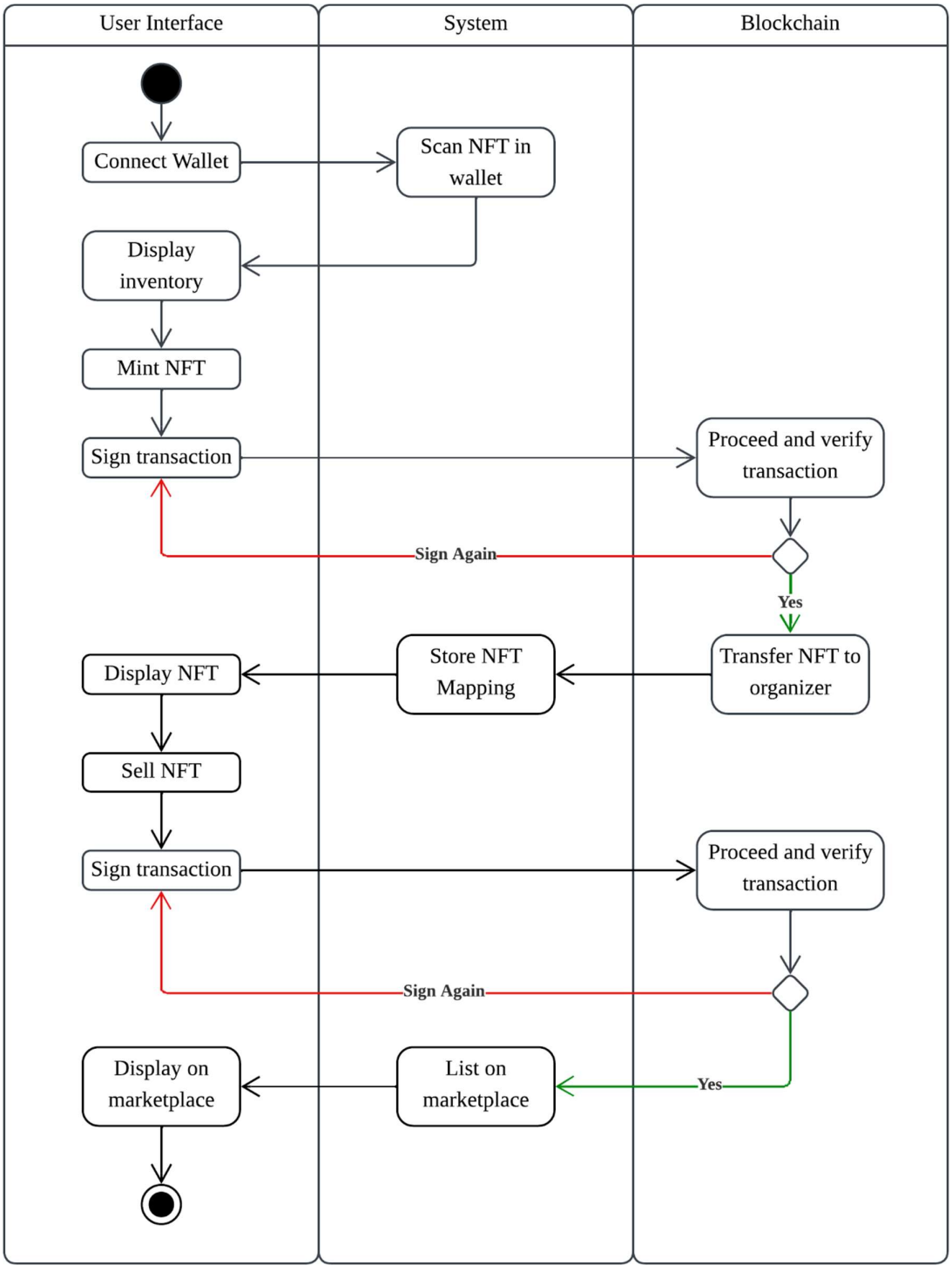


Figure 4.5 Activities diagram about mint and sell ticket

The mint and sell NFT operations are displayed in Figure 4.5. When the wallet is connected to the system, the NFT in the wallet will be identified by the system and will be shown in the user interface. The organizer is able to mint the NFT and validate the transaction. The blockchain will process and

validate the transaction; if it does not go through, the user will have to sign again. When the validation is successful, the blockchain will transfer the NFT to the organizer, and the transaction will be recorded in the system and displayed in the user interface. Subsequently, the organizer may sell the NFT by authentication the transaction. The blockchain will process and authenticate the sale transaction. If it is completed, the system will list the NFT on the marketplace and show it in the user interface.

4.7 Activity Diagram

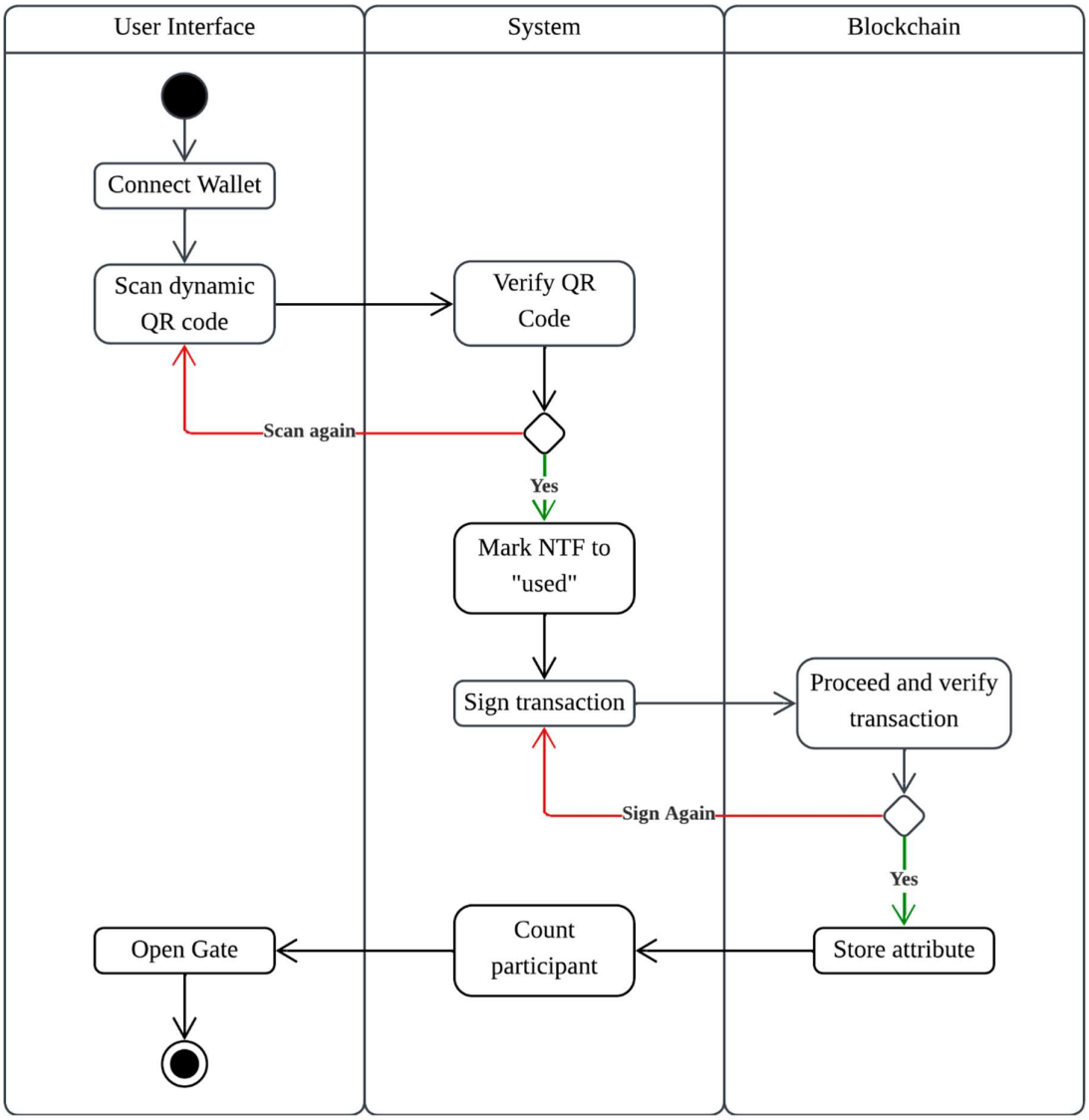


Figure 4.4 Activities diagram about ticket check-in

This illustration pertains to the QR codes checking and verification process. The participant uses his/her wallet to scan the QR code before the system checks its validity. In case of an invalid QR code, the participant will be required to scan again. Upon the successful check of the QR code, the NFT will

be marked as used. The transaction will then proceed to the blockchain frequency check. If this check fails, the QR code needs to be scanned again.

4.8 Summary

In this chapter, we introduce the system architecture diagrams that serve as the development guide. In the design phase, we use the Use Case diagram, Sequence diagrams, Class, and Activity diagrams.

The Use Case diagram illustrates how users interact with the system, which includes functionalities like minting, purchasing, and reselling of the NFT tickets, check-in to the event, and management of funds. The Sequence diagram shows the sequential technical implementation required to create the event with different steps performed on the website, off-chain database, and blockchain.

Additionally, the Class diagram reflects the static structure of the system by differentiating between off-chain like User, Event and Web3BackendService components and on-chain components like TicketSmartContract and TicketMetaData. Lastly, the Activity diagrams delineate the conditional workflows for purchasing and listing the tickets on the secondary market, followed by the QR code check-in process so as to ensure secure transactions on the blockchain.

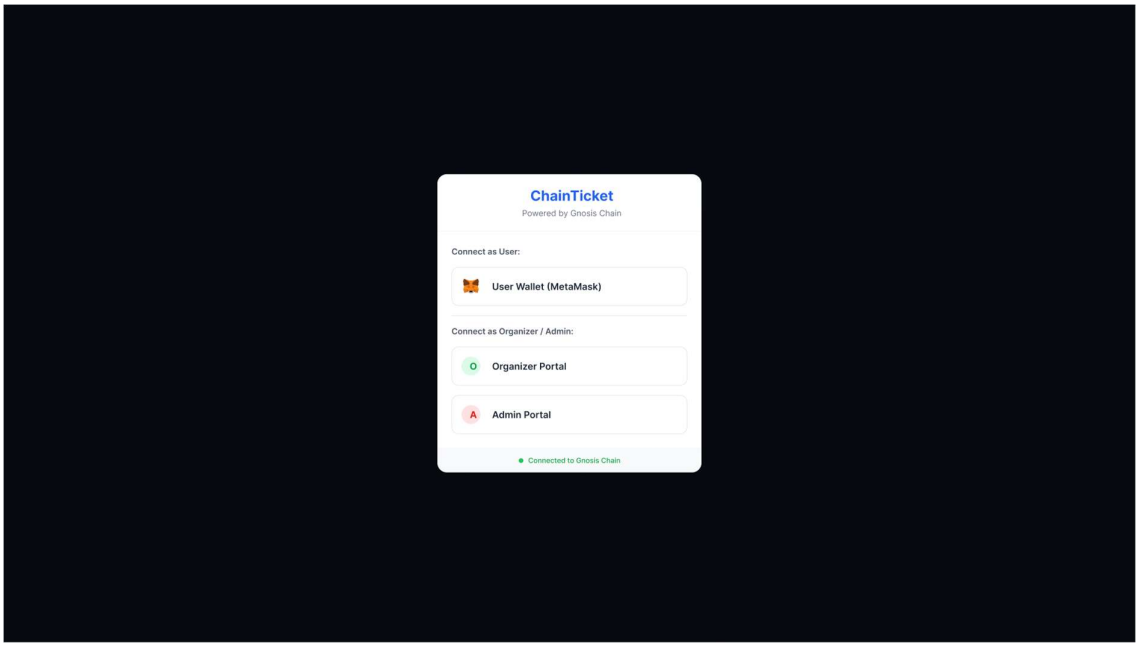
Chapter 5: Implementation (FYP I) prototype

5.1 Introduction

The ChainTicket system prototype will be demonstrated in this chapter using a dummy page and description to give a visual depiction of how the system operates.

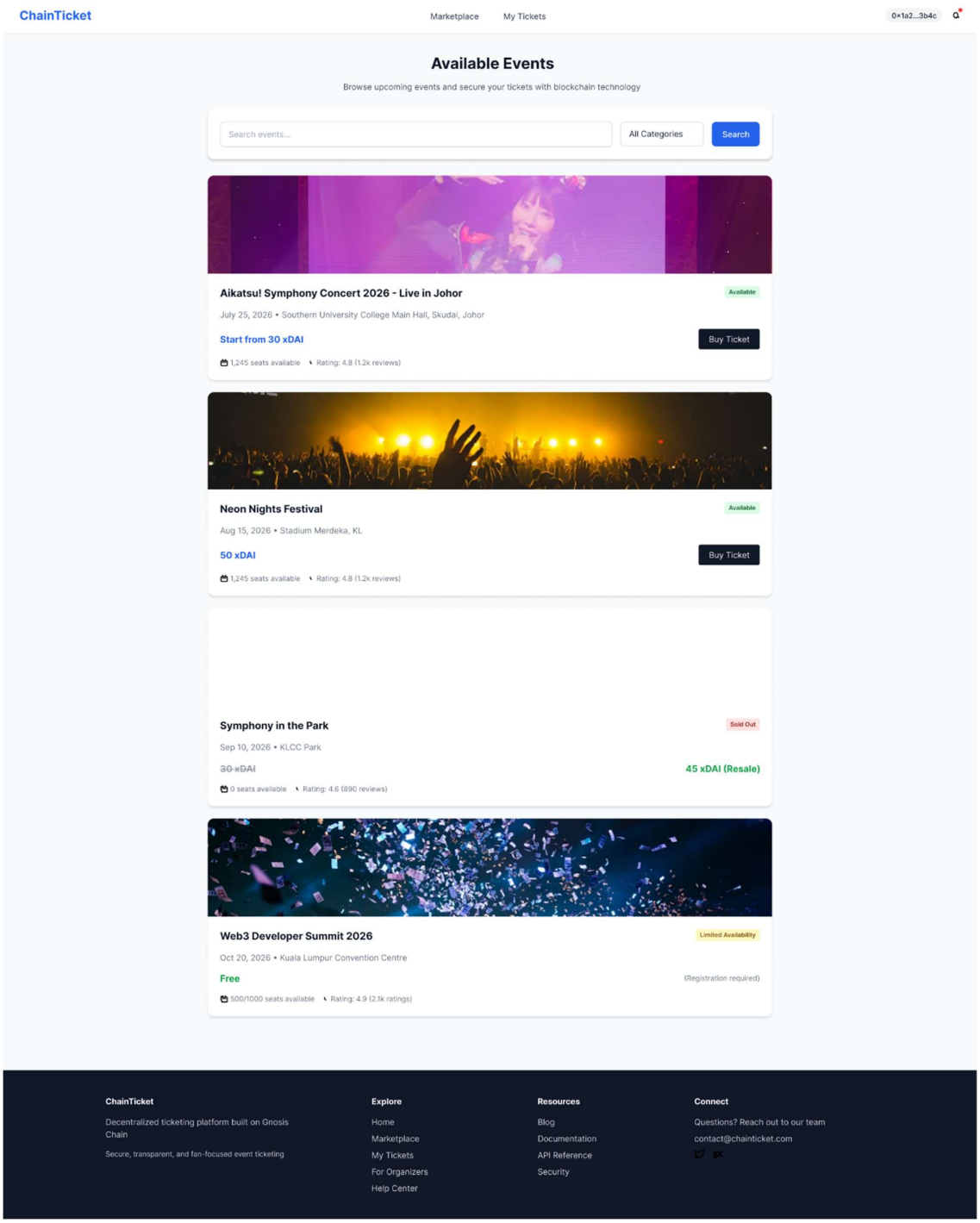
5.2 Prototype

5.2.1 Connect Wallet



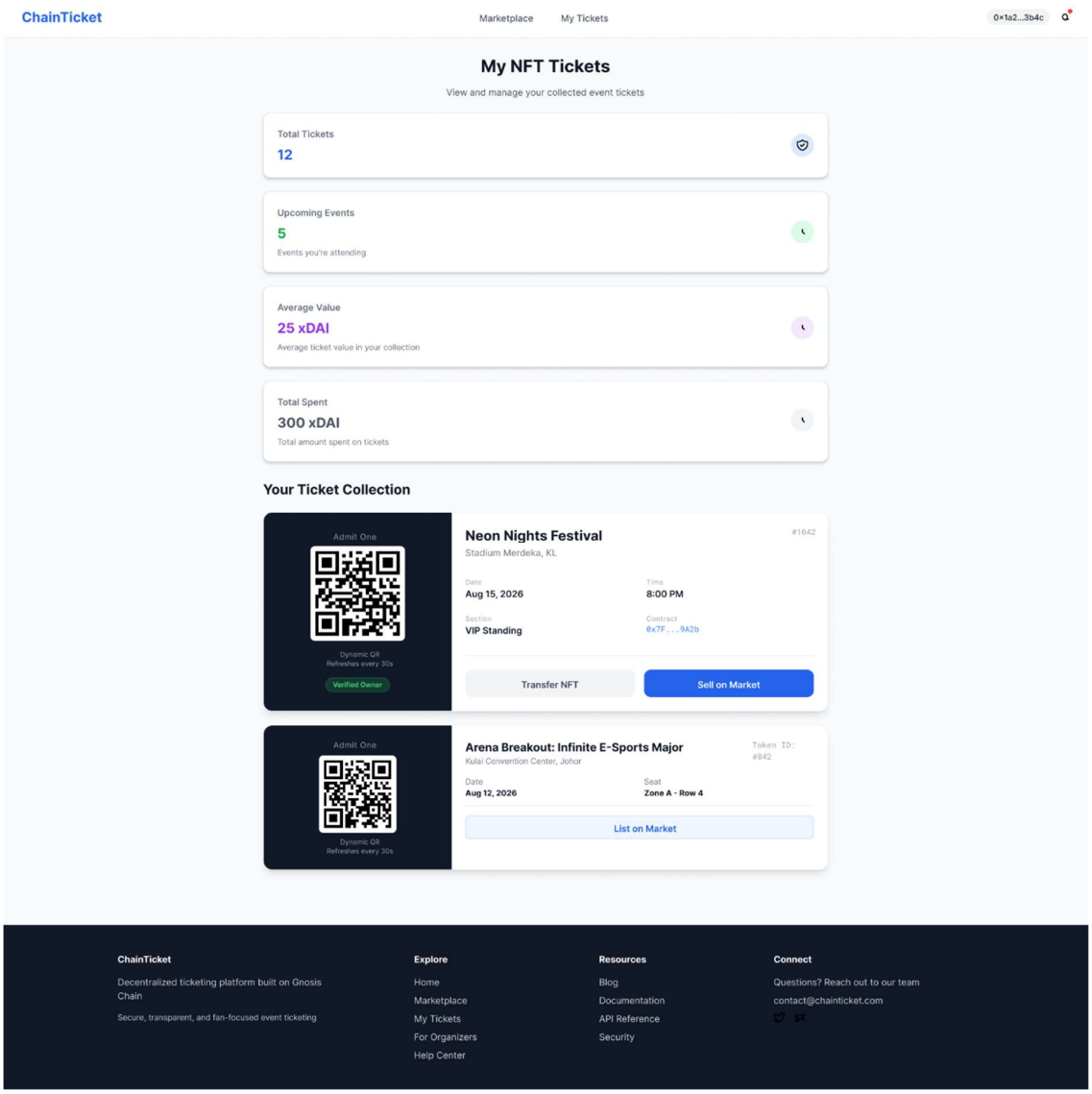
This login screen allows users to log in into the platform using their Web3 identity. It features two modes of connecting: regular user with MetaMask, organizer, and admin. The status indicator tells users whether they are connected to the Gnosis Chain.

5.2.2 Marketplace (Client)



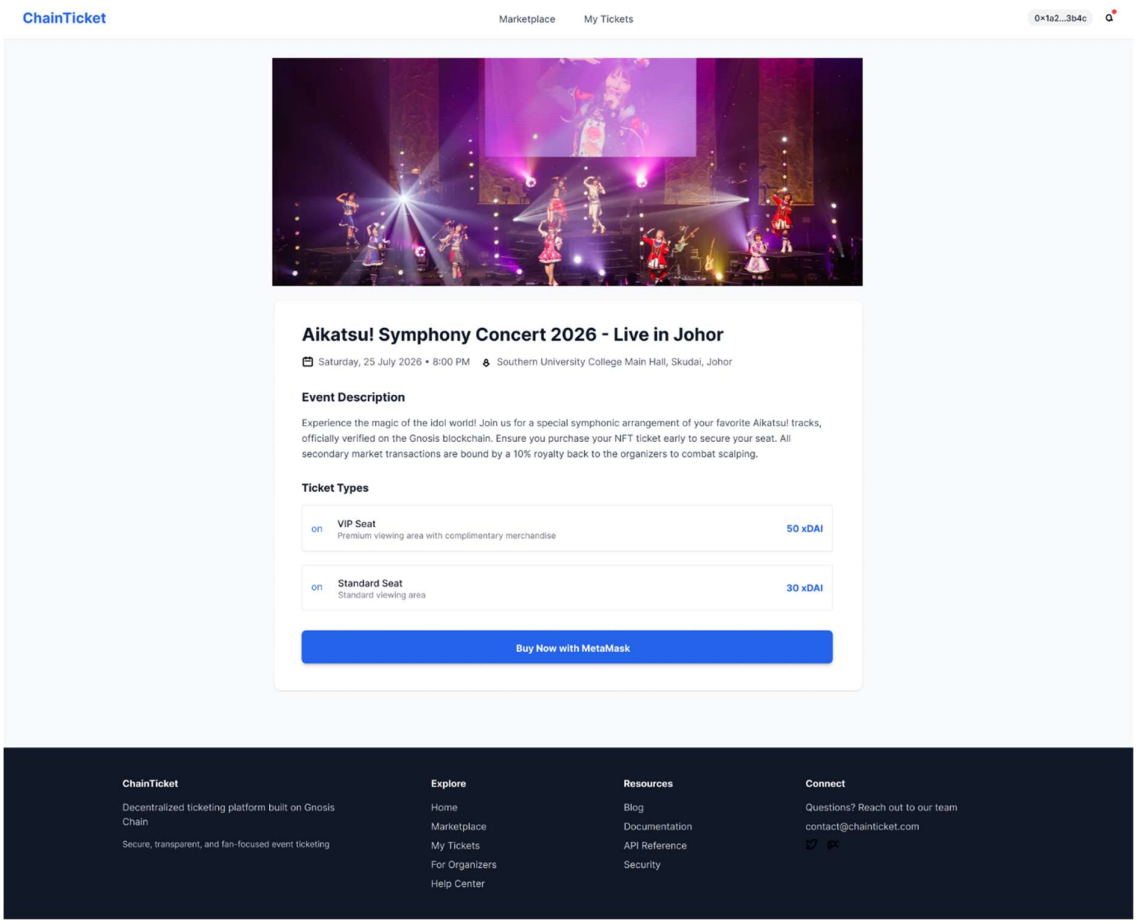
This page permits users to browse, search, and filter events. It displays event cards with data on ticket prices in xDai, availability of seats, and ratings. Users may know whether tickets are available, sold out, or resold before they decide to click to purchase.

5.2.3 My Ticket (Client)



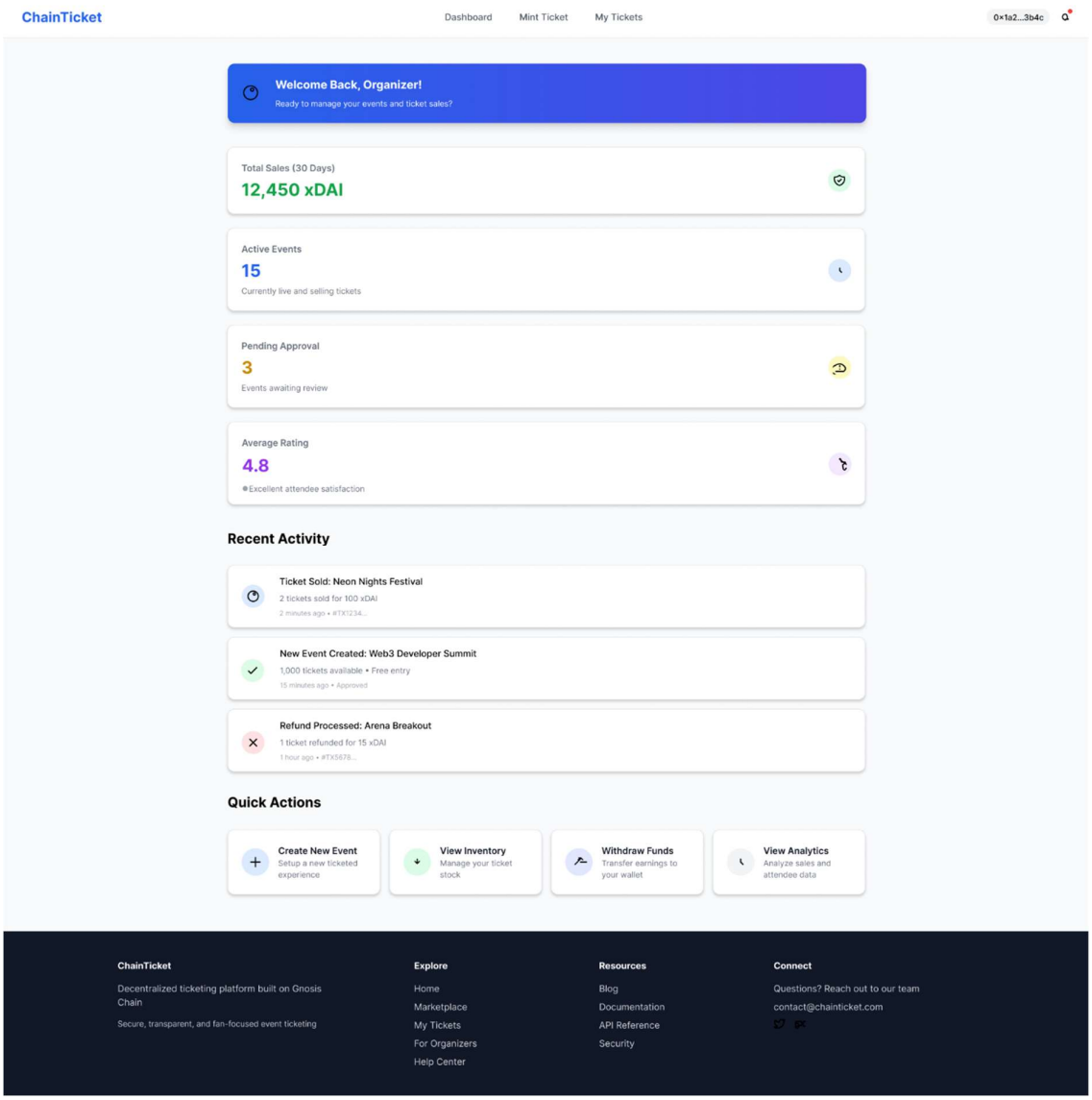
This user's personal area is dedicated to the set of NFT tickets in possession and shows summary stats, including upcoming events and total xDAI spent. The ticket collection has QR codes which are time-limited and updated every 30 seconds

5.2.4 Ticket information



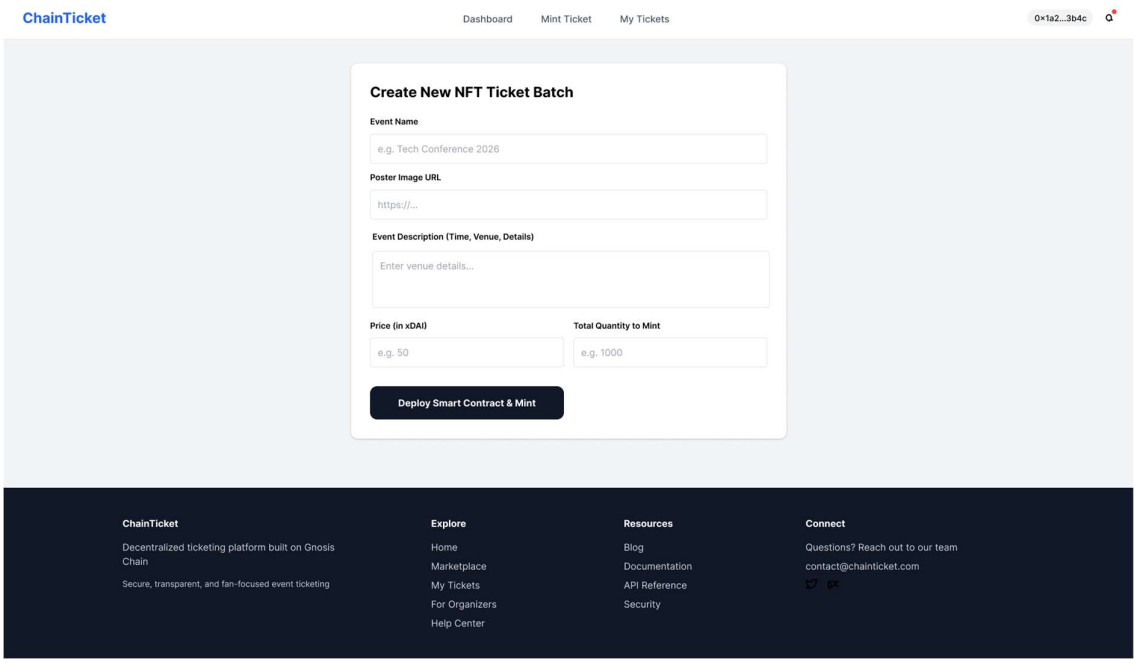
This page shows the complete information about a particular event, including date, time, location, and description. It also shows different types of tickets, such as VIP or standard, including prices in xDAI. The purchase is made using the “Buy Now via MetaMask” button.

5.2.5 Event dashboard (organizer)



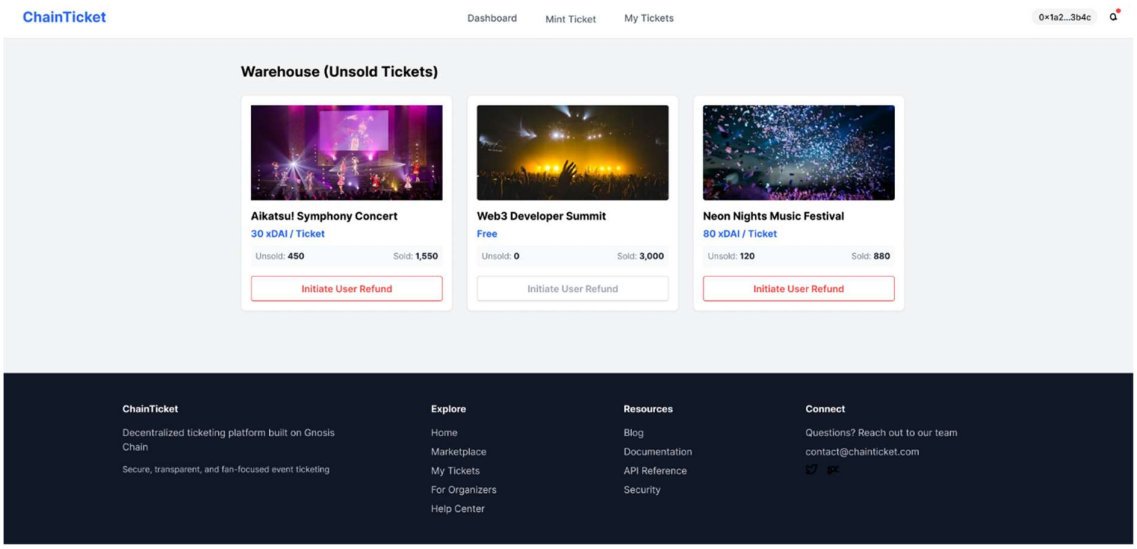
This pages serves as the main area for event organizers to keep an eye on their ticketing operations. It provides generalized data about general ticket sales in xDAI, ongoing events, and average levels of galas performed successfully by any ticket buyers. There is a recent activity report as well as simple control icons that help organization with event arrangement, supply management, and money transfer.

5.2.6 Mint Ticket (Organizer)



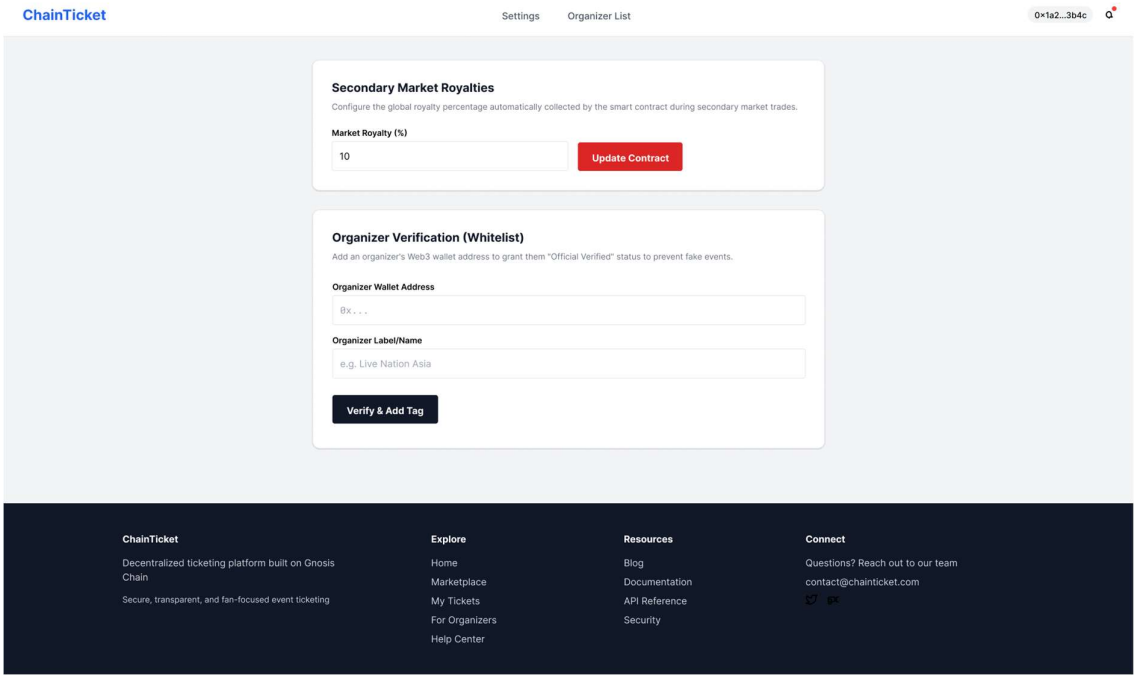
This page enables event organizers to prepare a fresh batch of NFT tickets originating from some upcoming events. The user needs to introduce the event name and its poster address, event description, a ticket price expressed in xDAI, and the total number of sold tickets. Once the form is submitted, the NFT ticket gets prepared for block generating.

5.2.7 My Ticket/Inventory (Organizer)



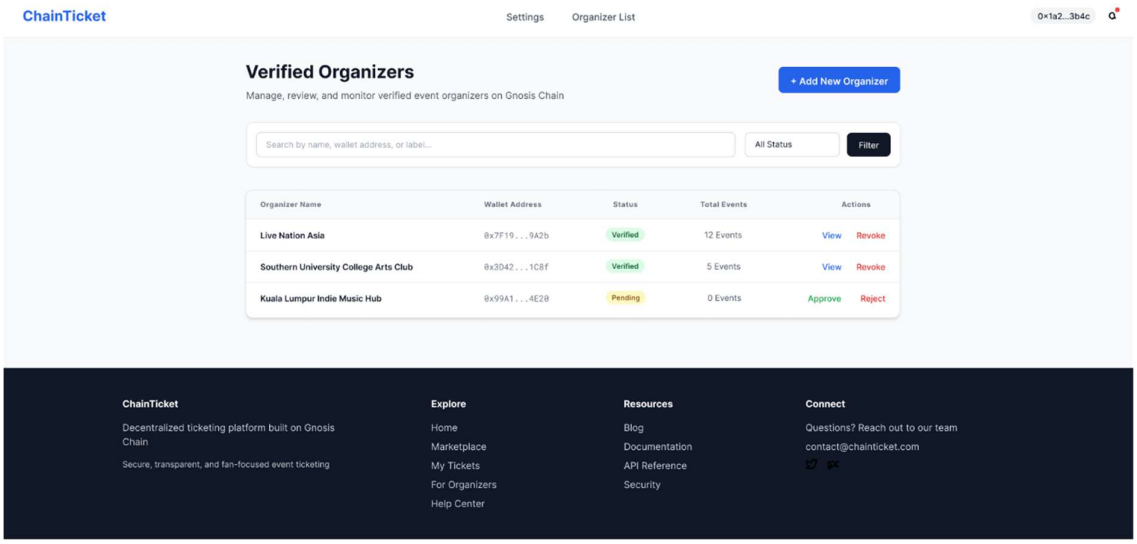
This stock management area reveals full information about already generated (but not sold) tickets and the page also displays cards with data about number of tickets sold and number of tickets that are still in stock. Event organizers can make usage of this tool to get refunds processed for certain events if required.

5.2.8 Setting (Admin)



This admin page is used to set up various settings for the system. This includes options for the secondary market royalty percentage that the smart contract will take automatically. One feature of this is the whitelist that can be used to confirm and approve anyone who is an official organizer.

5.2.9 Organizer List(Admin)



This feature of the admin page allows an admin to keep a track of the authorized organizers in the database. The organizer database allows for an admin to see relevant information about each organizer, including their name and wallet address, and how many events they have hosted. There are functionalities that let admins approve organizers who are still pending, as well as revoke the ones that have already gained access.

5.3 Summary

Decentralized ticketing platform's user interface is developed to present an effortless experience in Web2.5 model for three differentiated roles that are attendees, organizers and administrators. The interfaces act on Gnosis Chain and conceal intricate blockchain operations for safe transactions.

Attendees get access to an easy-to-use marketplace for availing upcoming events, get a chance to buy tickets through xDAI and monitor NFT collections. Moreover, the dashboard contains a changing QR code that refreshes each 30 seconds for enhancing safety of the event's attendance.

Organizers enjoy the benefits of a complete managing platform. They can access analytics in real-time, get a direct access to smart contracts to mint tickets and take inventory of unused stock.The administrative module of the platform guarantees safety of the platform's management. In the admin module, administrators can use settings to globalize parameters such as automatic market fees levied on organizing events. The roles are designed to provide a safe platform for everyone involved and working in the event industry.

Chapter 6: Conclusion

6.1 Introduction

Chapter 6 summarizes all the ideas gathered during the FYP1. It is a conclusion where the information gathered can serve as the basis for FYP2. Section 6.2 reviews the achievements and the knowledge obtained so far in the project. Section 6.3 discusses system limitations encountered during systems design and prototyping phases. Section 6.4 indicates the future improvements based on the limitations mentioned above in order to improve the quality of the project in FYP2. Lastly, Section 6.5 gives a summary of the whole chapter and the project in general.

6.2 Achievements

I. Blockchain Integration and Web3 Architecture

- Successfully designed a decentralized Web2.5 ticketing architecture utilizing the Gnosis Chain. Gained in-depth knowledge of developing smart contracts using Solidity and the Foundry framework to mathematically enforce secondary market pricing and automated royalty splits.

II. Anti-Fraud Security Implementation

- Learned and conceptualized the mechanics of Dynamic QR code authentication. Achieved a comprehensive design that uses time-sensitive cryptographic tokens to prevent ticket duplication and screenshots, solving major vulnerabilities in legacy e-ticketing systems.

III. System Design and Prototyping

- Successfully mapped out complex user workflows through comprehensive UML diagrams (Use Case, Sequence, Class, and Activity diagrams) and developed a functional UI/UX prototype catering to Users, Organizers, and Admins.

IV. Hybrid Data Management

- Acquired a practical understanding of balancing on-chain and off-chain data storage. Learned to leverage Firebase for user profiles and real-time analytics, alongside Pinata (IPFS) for decentralized metadata storage, ensuring low gas fees and high system efficiency.

6.3 Limitation

I. Web3 Onboarding Barrier

- The current system design heavily relies on users connecting a Web3 wallet like MetaMask or OKX Web3 Wallet to purchase and store their NFT tickets. This presents a steep learning curve and accessibility barrier for non-technical users who are unfamiliar with blockchain technology, private keys, and decentralized applications.

II. Network Dependency and Gas Fees

- Although deployed on the highly efficient Gnosis Chain, the system is still subject to the inherent constraints of public blockchain networks. Users must possess native cryptocurrency

(xDAI) to pay for transaction gas fees. This complicates the checkout process compared to traditional platforms where fiat currency is standard.

III. Offline Verification Constraints

- The Dynamic QR code verification terminal requires a continuous, stable internet connection at the event venue to interact with the blockchain state in real-time. The current prototype lacks a robust offline synchronization mode, which could cause major crowd bottlenecks if the venue experiences unexpected network connectivity issues.

6.4 Future enhancements

I. Account Abstraction and Social Login

- To lower the Web3 entry barrier, future iterations will integrate Account Abstraction (ERC-4337) or Web3 Auth solutions. This will allow users to create and manage non-custodial wallets using standard email or social media accounts, eliminating the strict requirement for external browser extensions like MetaMask.

II. Fiat Payment Gateway Integration

- The system will be enhanced by implementing a fiat-to-crypto payment gateway like Visa Card payment integration. This feature will automatically handle currency conversion and gas fee sponsorship (meta-transactions) in the background, allowing users to purchase NFT tickets seamlessly using traditional credit and debit cards.

III. Offline-Capable Verification Modules

- Future development will enhance the verification terminal with a localized edge-caching system or cryptographic zero-knowledge proofs. This will allow for the offline authentication of Dynamic QR codes, ensuring smooth crowd entry even in remote venues with poor internet connectivity, automatically syncing the check-in data to the blockchain once the network is restored.

6.5 Summary

In summary, the Web-Based Ticket System created by ChainTicket offers a groundbreaking decentralized approach to dealing with the challenges of ticket scalping, monopoly control, and duplication fraud in the live events sector. Apart from the theoretical framework, UML designs, user interface prototypes, and hybrid Web 2.5 architecture, FYP1 shows certain limitations that the team faces, including the barriers of usability of Web 3.0 technology and continuous reliance on the network. Thus, ChainTicket will be able to provide a secure and transparent digital ticketing system.